

LING2136 / 4136

**Advanced Statistical Methods for
Language Students**

Session-09: RDFs

Lecturer: Timo Roettger



ONE QUESTION MANY ANSWERS

On Researcher Degrees of
Freedom and Vague Theories

Timo Roettger
University of Oslo



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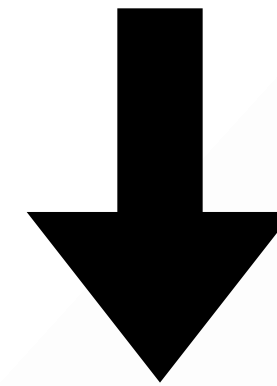


**COMMUNICATION IS
COOPERATIVE**

BE AS INFORMATIVE AS YOU CAN...AND NO MORE

Grice (1975)

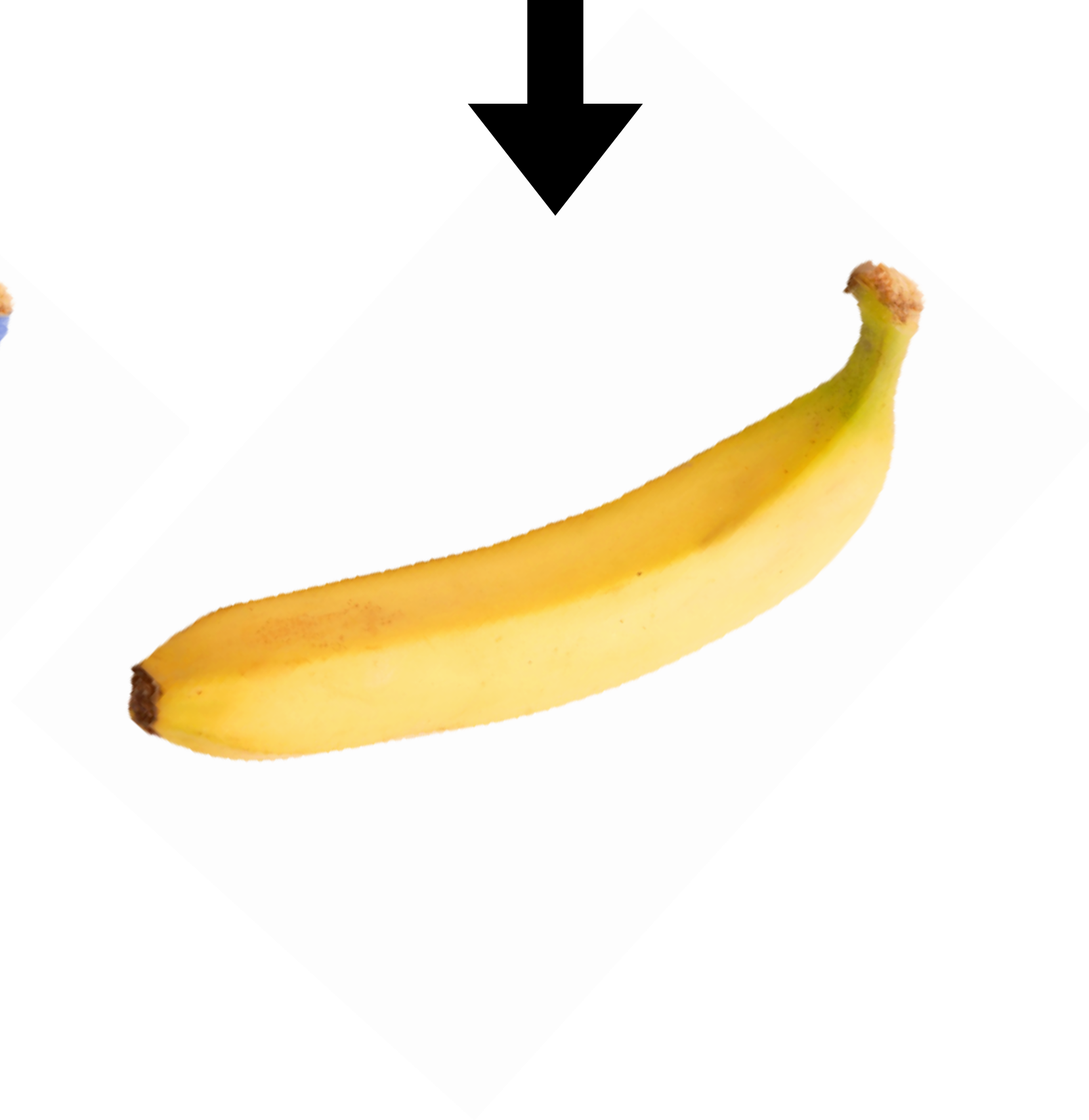
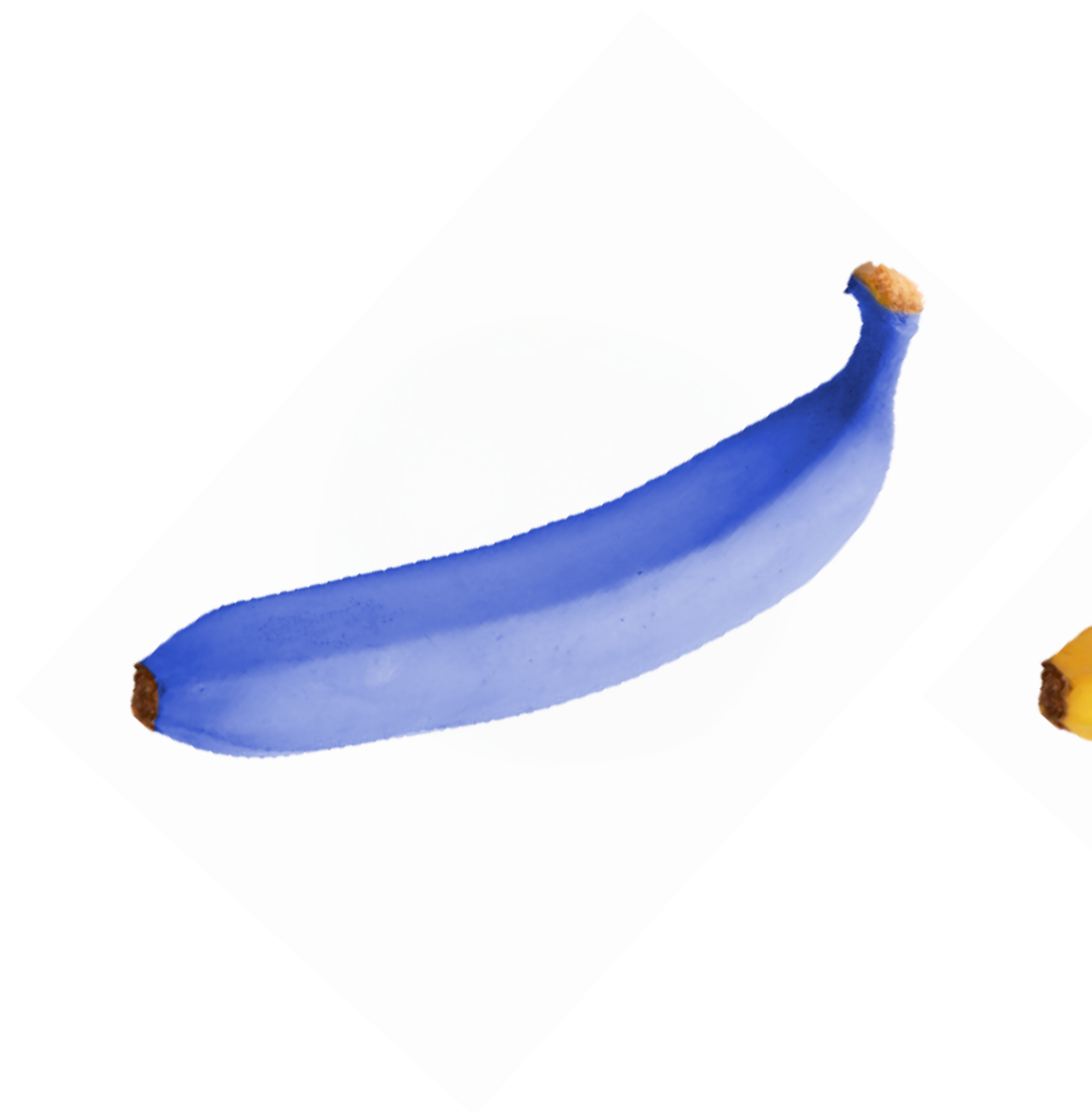
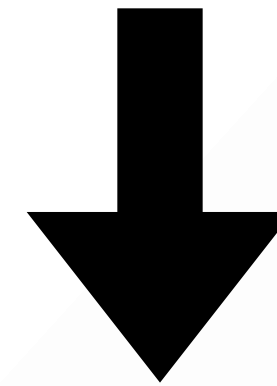
What object is marked by an arrow?



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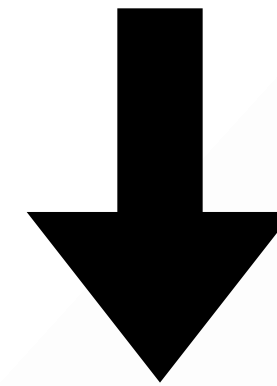
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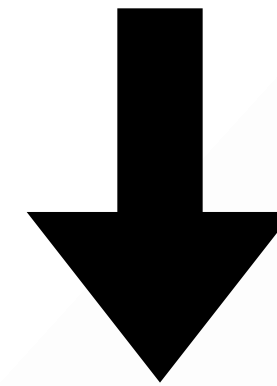
What object is marked by an arrow?



BUT WE OFTEN USE REDUNDANT MODIFIERS

e.g. Sedivy (2003), Westerbeek et al. (2015), Rubio-Fernandez (2016)

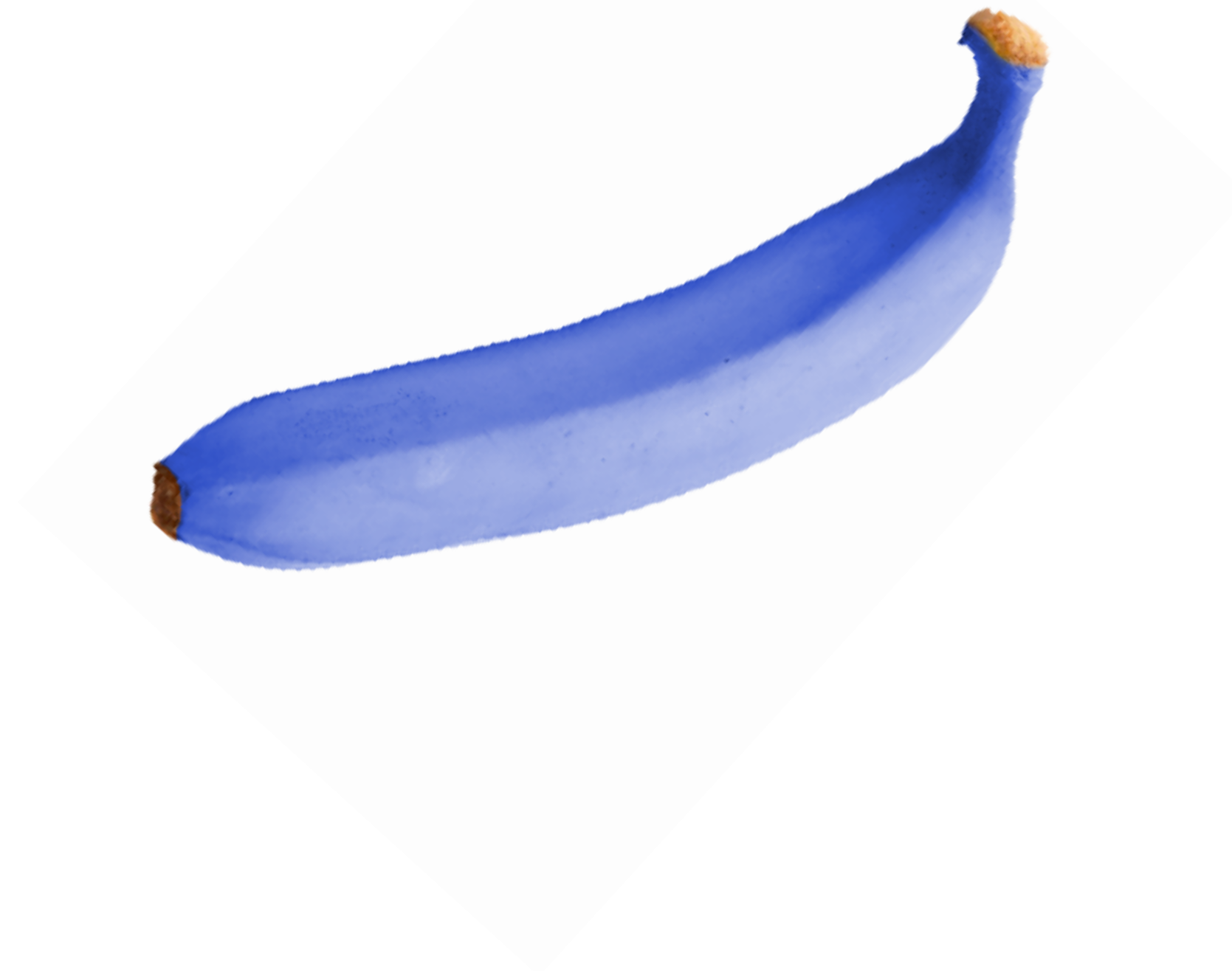
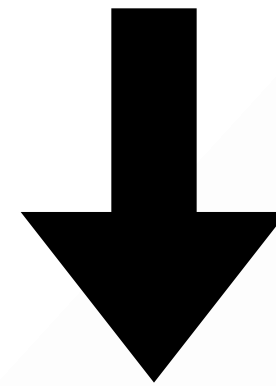
What object is marked by an arrow?



...MORE OFTEN WHEN THE MODIFIER IS ATYPICAL

Degen et al. (2023)

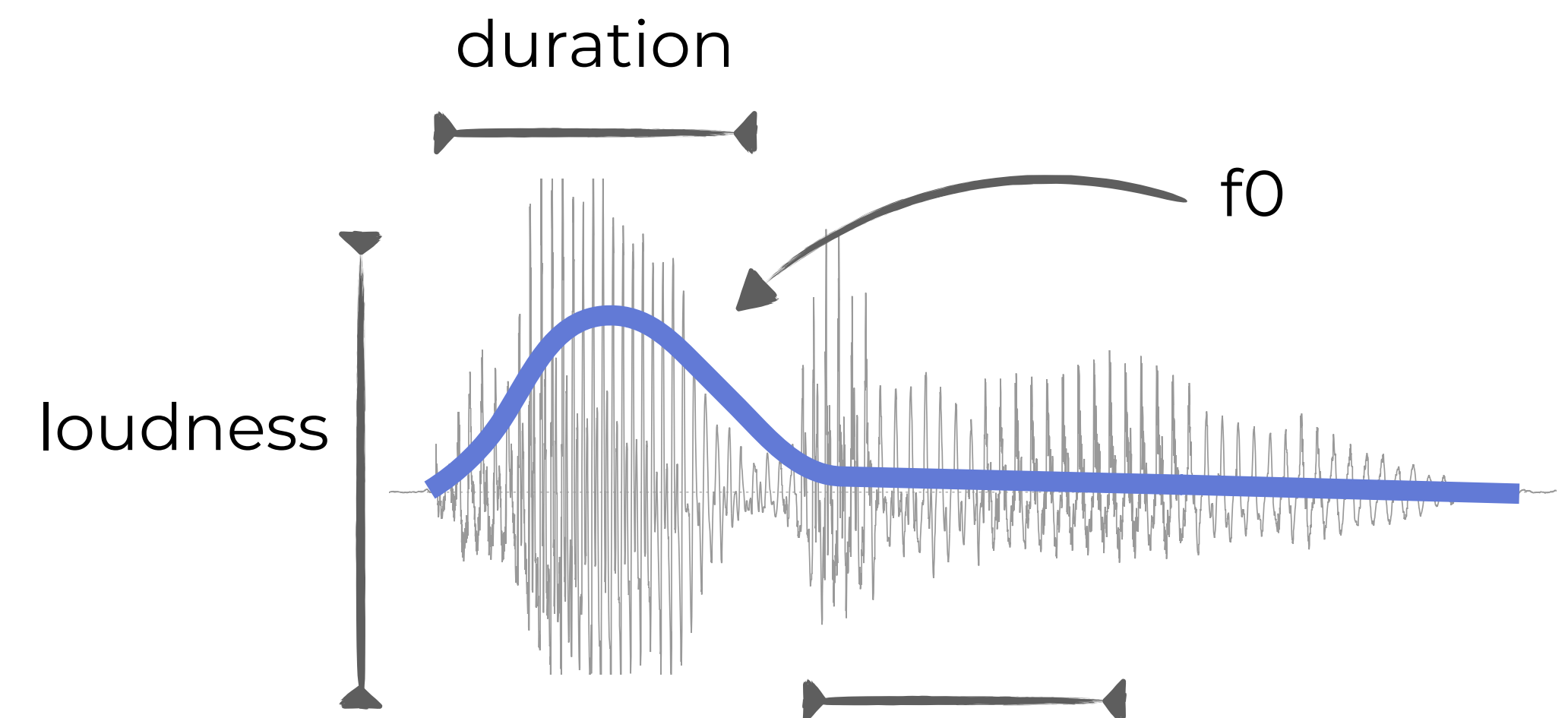
What object is marked by an arrow?



BUT DOES IT MATTER HOW WE SAY IT

Prosody can signal

information structure	e.g. Mücke & Grice (2014)
information status	e.g. Baumann & Grice (2006)
sentence modality	e.g. Hirst & Di Christo (1998)
predictability & importance	e.g. Turnbull (2017) Cole & Roettger (2019)

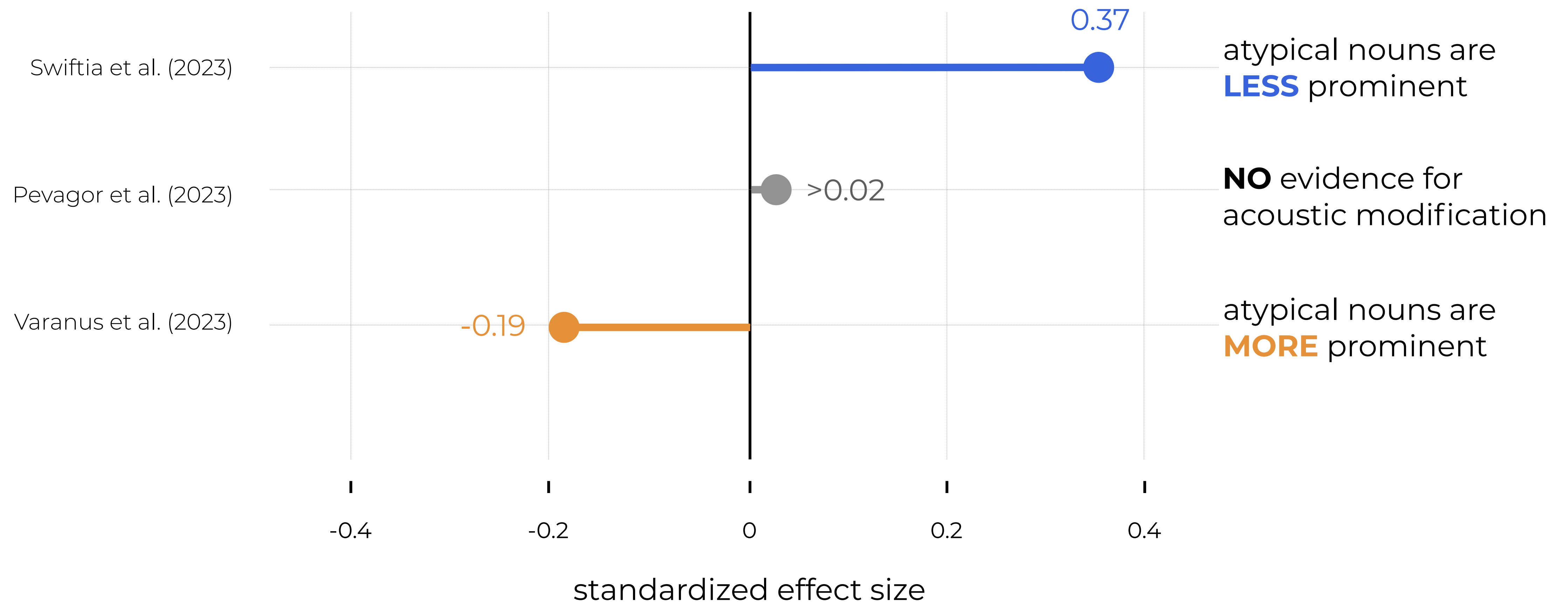


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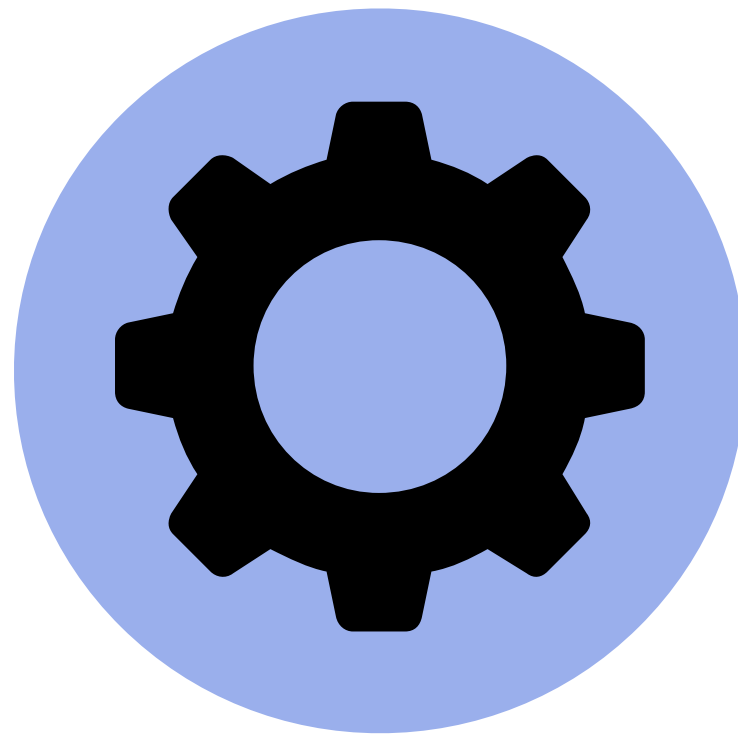
Do speakers acoustically modify utterances to signal atypical word combinations?

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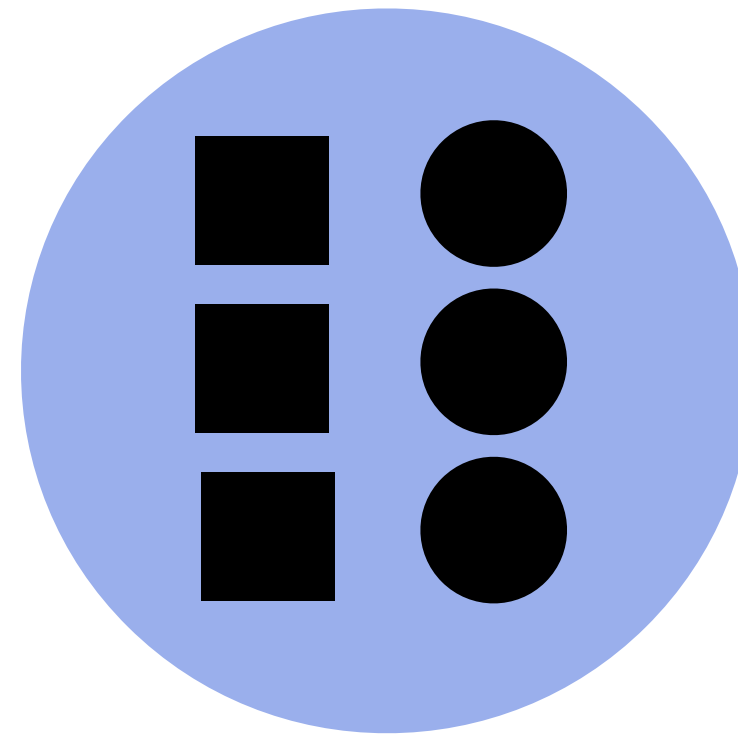
3 RESEARCH TEAMS, 3 DIFFERENT ANSWERS



NOT SURPRISING, IS IT? **STUDIES** DIFFER



in their design



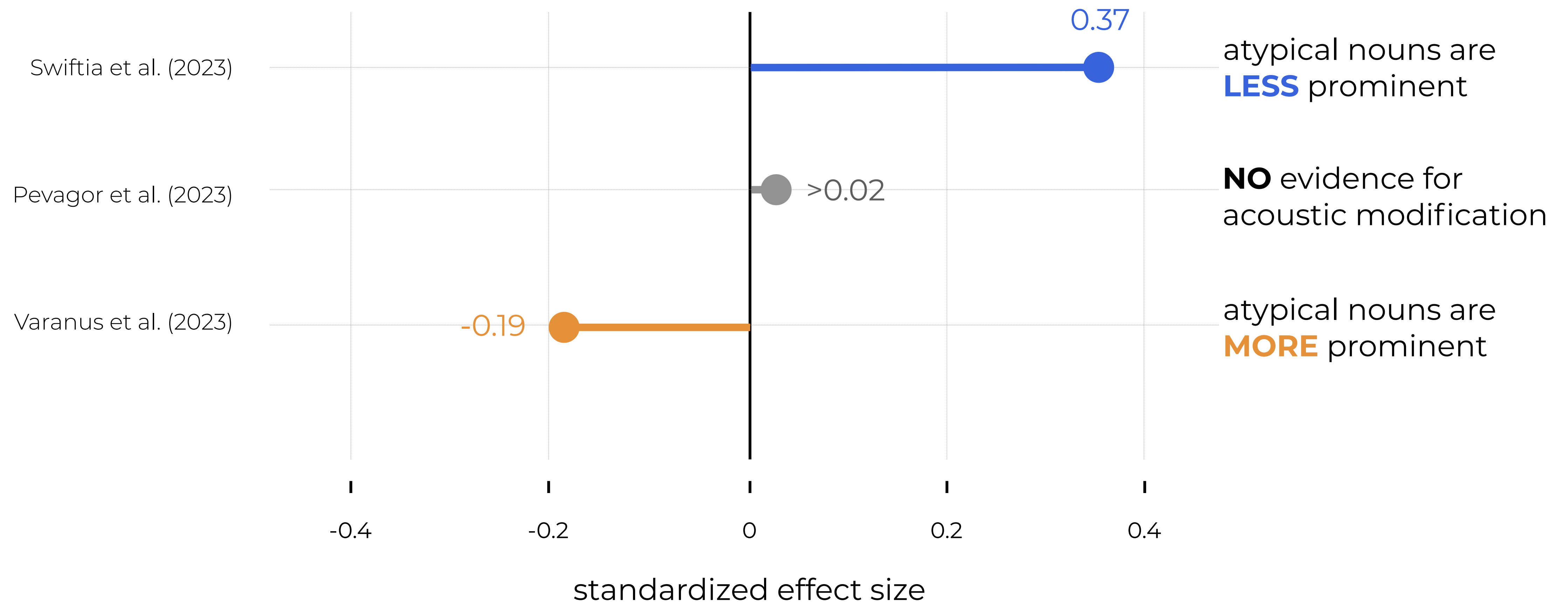
in their stimuli



in their sample

Do speakers acoustically modify utterances to signal atypical word combinations?

3 RESEARCH TEAMS, 3 DIFFERENT ANSWERS



STILL NOT SURPRISING BECAUSE **RESEARCHERS** DIFFER



quality of analysis



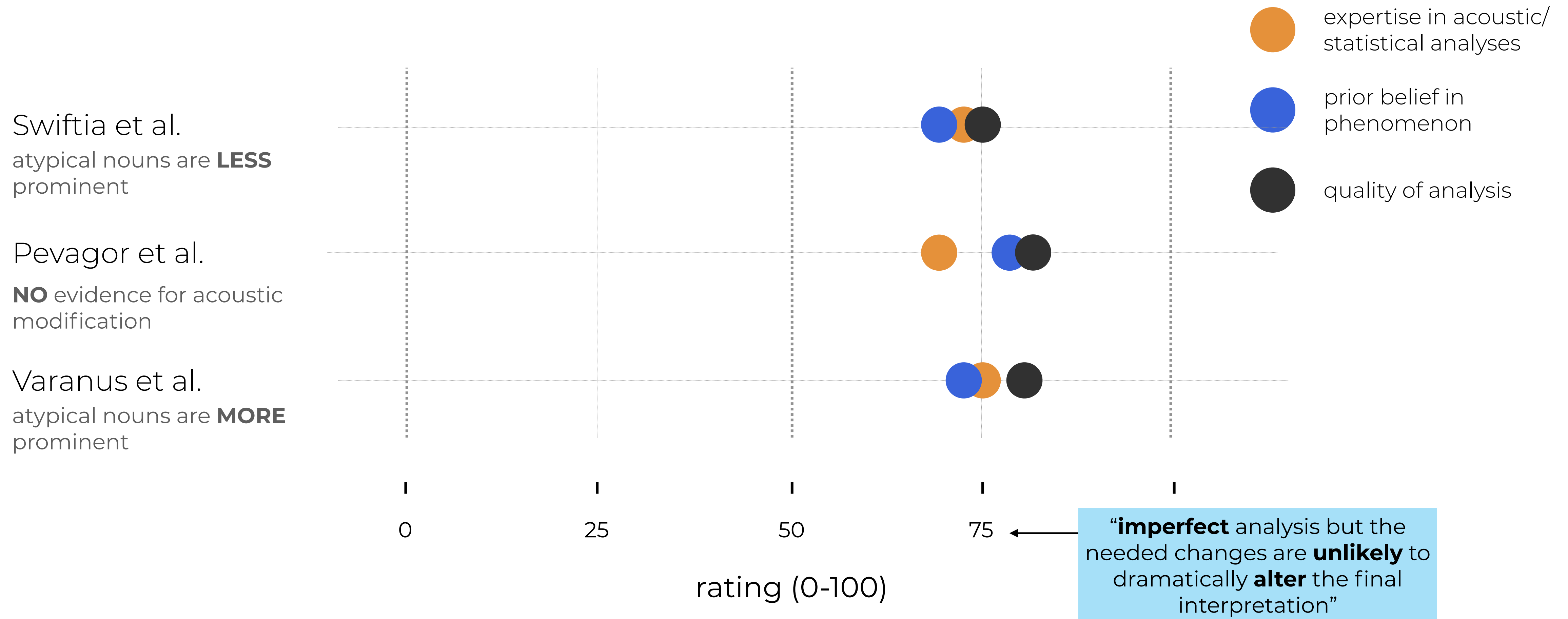
experience of analyst



bias of analyst

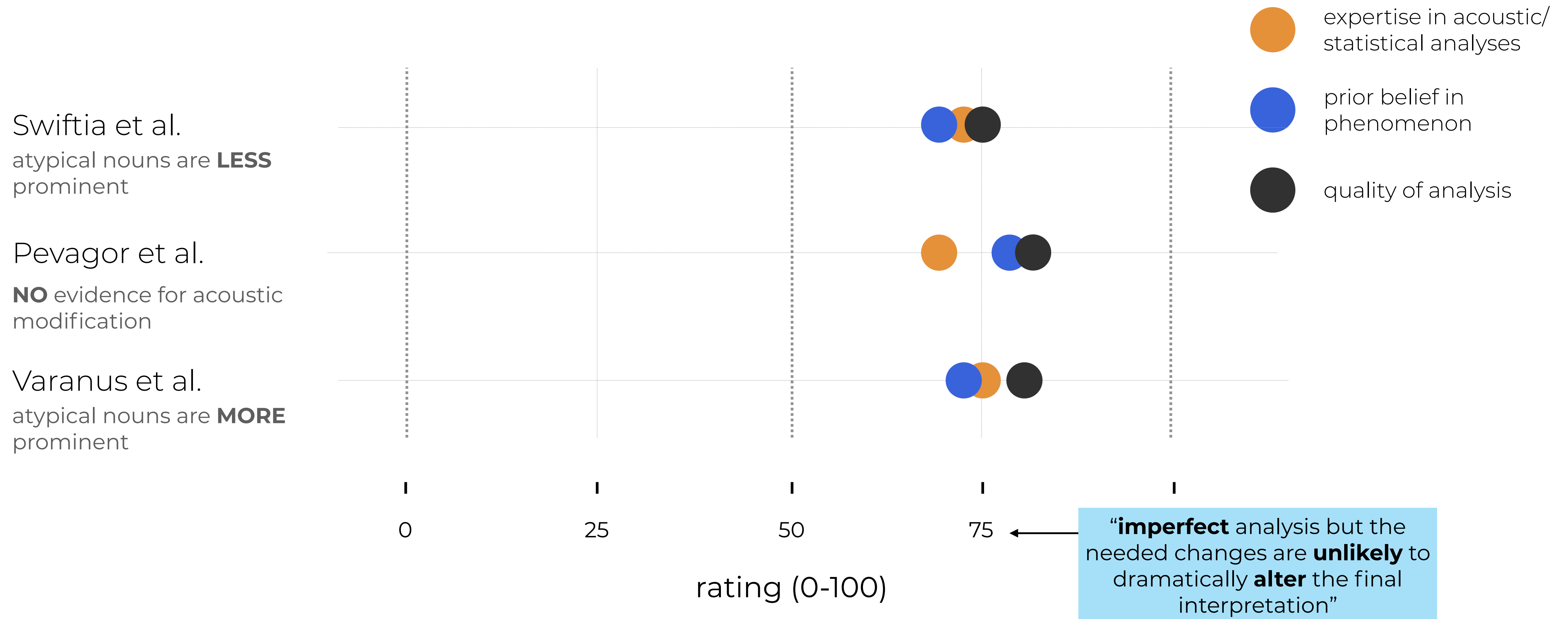
Can we explain the discrepancies with quality, expertise or bias?

NO, THEY ARE COMPETENT AND EQUALLY BIASED



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NO, THEY ARE COMPETENT AND EQUALLY BIASED





MANY SPEECH ANALYSIS

Coretta, Casillas, [153 more authors], **Roettger** (2023). Multidimensional signals and analytic flexibility: Estimating degrees of freedom in human speech analyses. *Advances in Methods and Practices in Psychological Science*.

35%

duration

36%

f0

13%

formants

14%

intensity

other

what?



what?



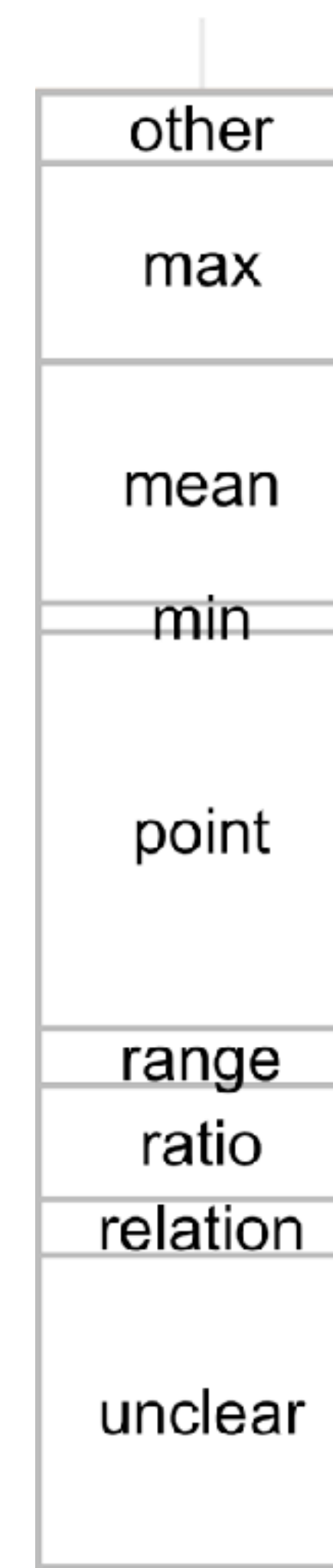
where?



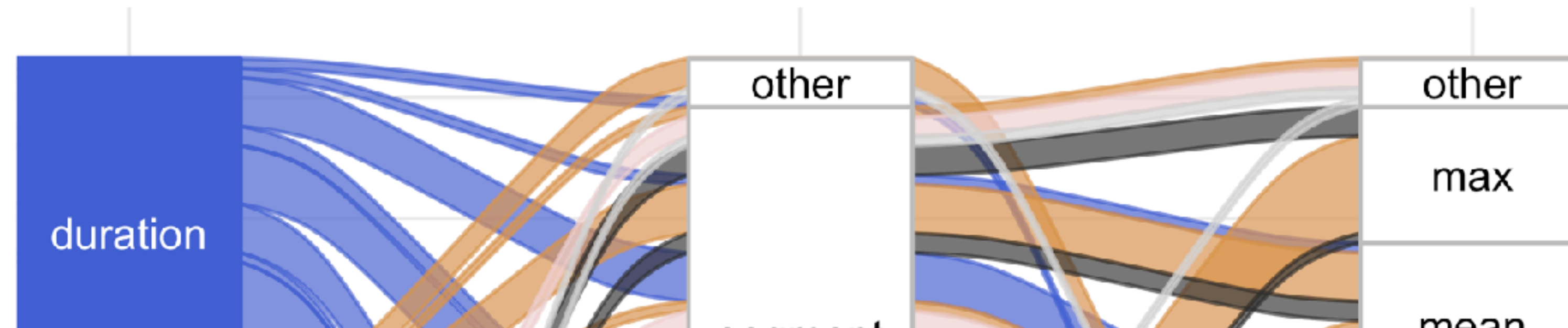
what?



where?

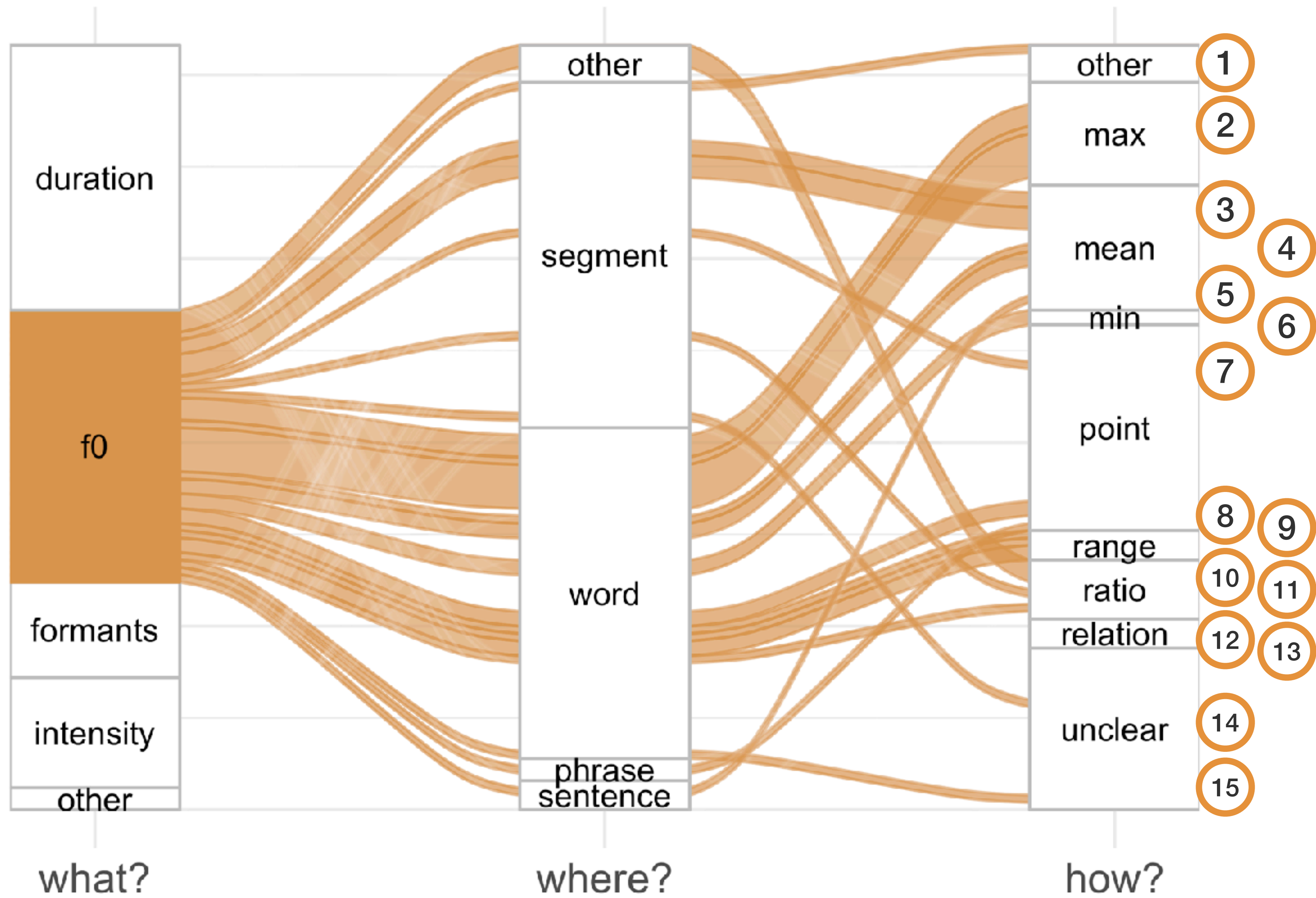


how?



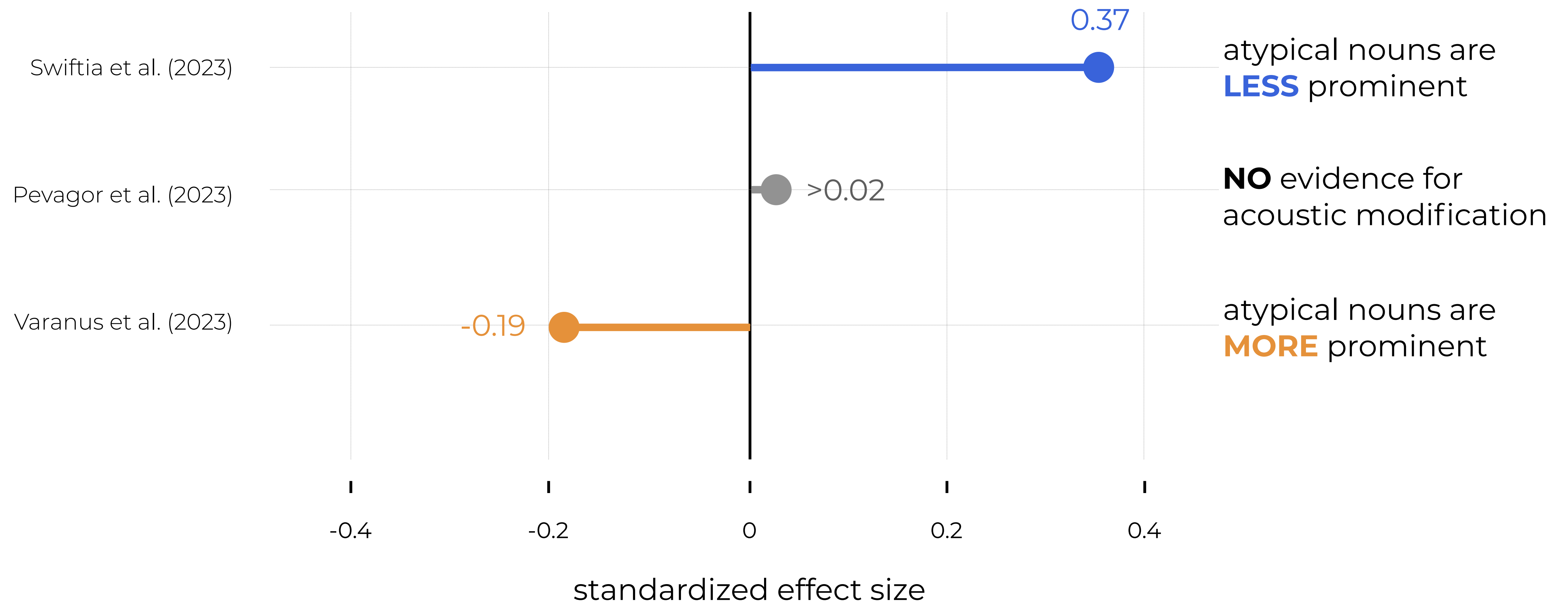
EVERY ACOUSTIC
ANALYSIS WAS **UNIQUE**

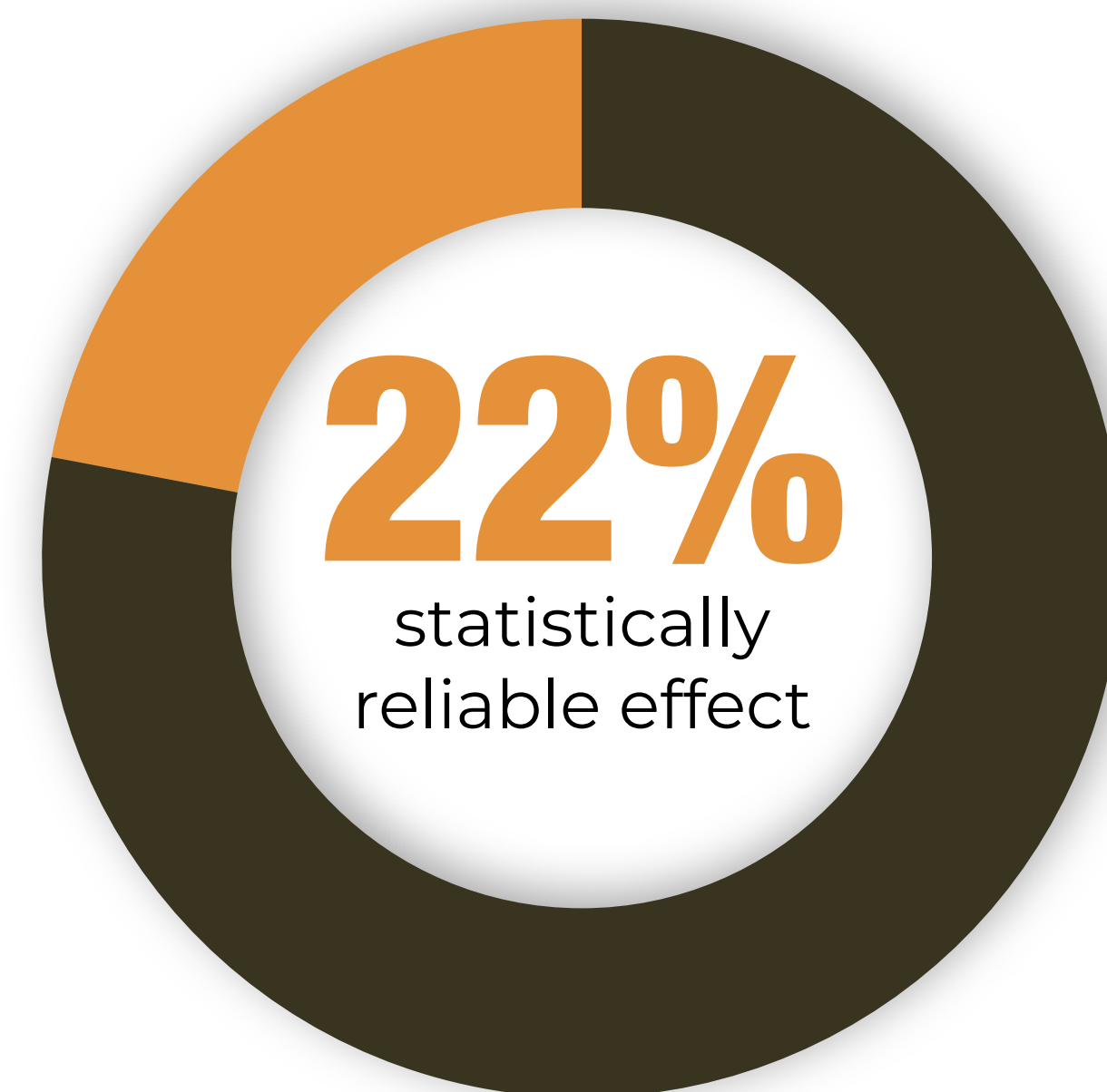
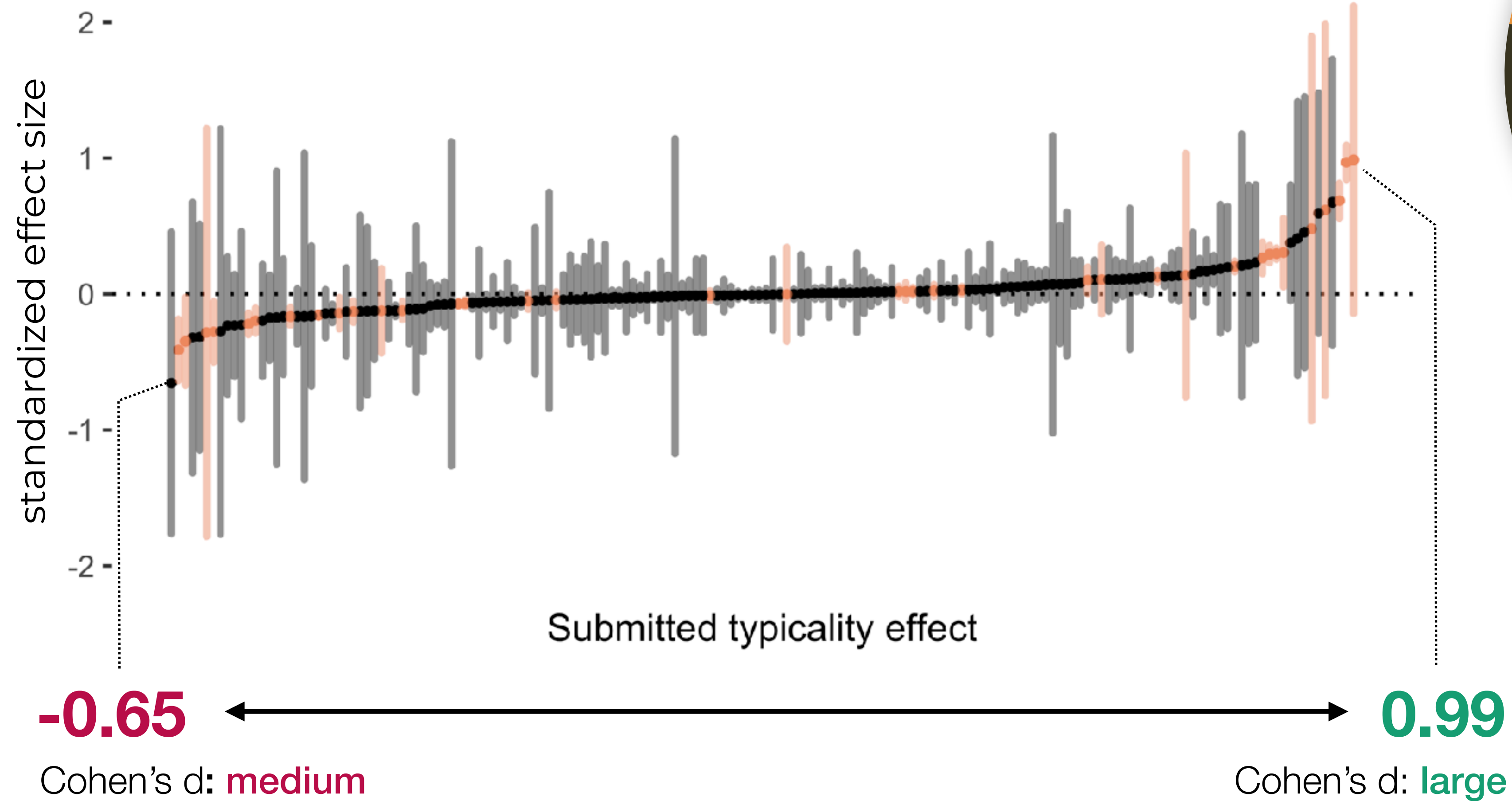


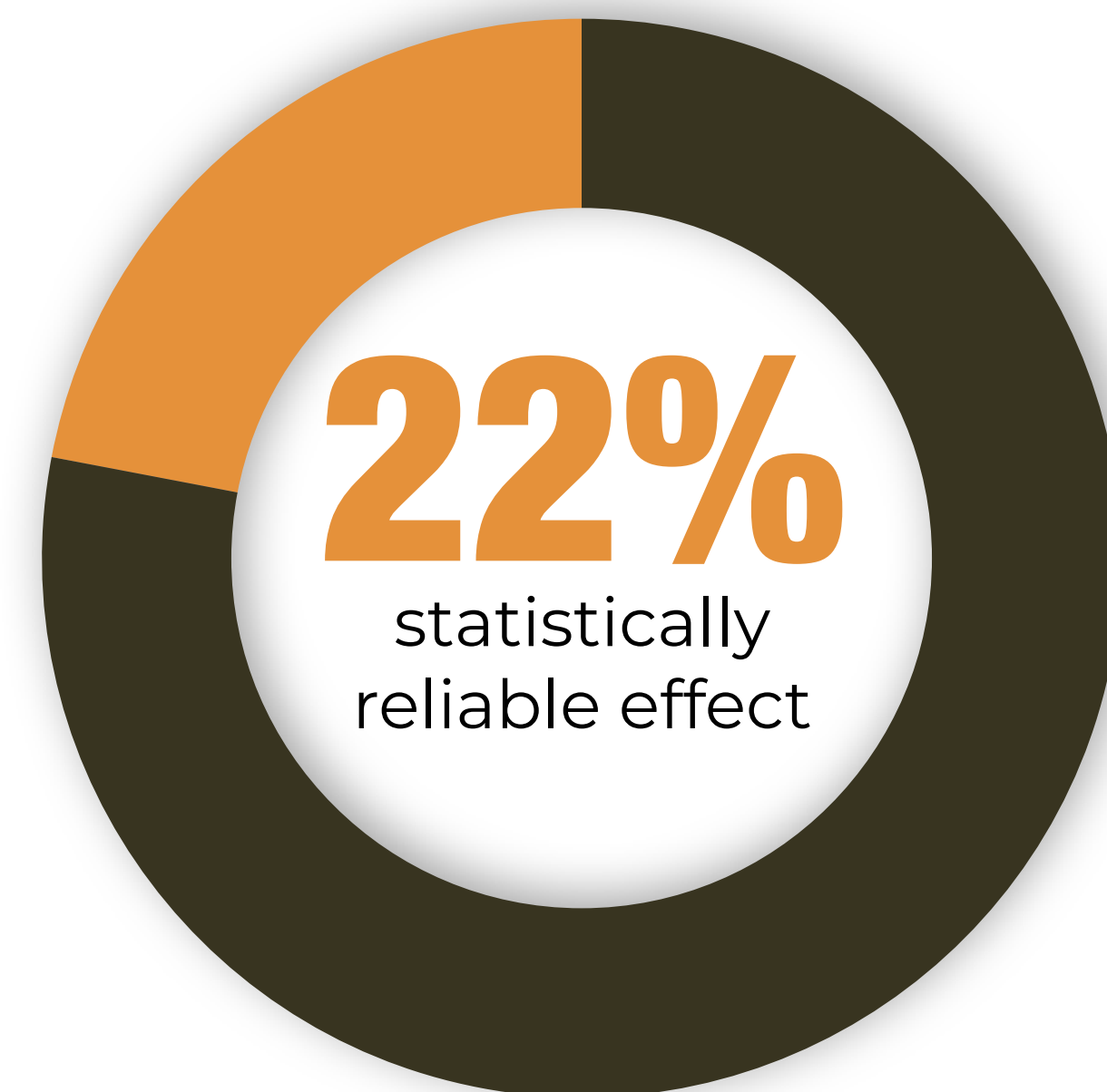
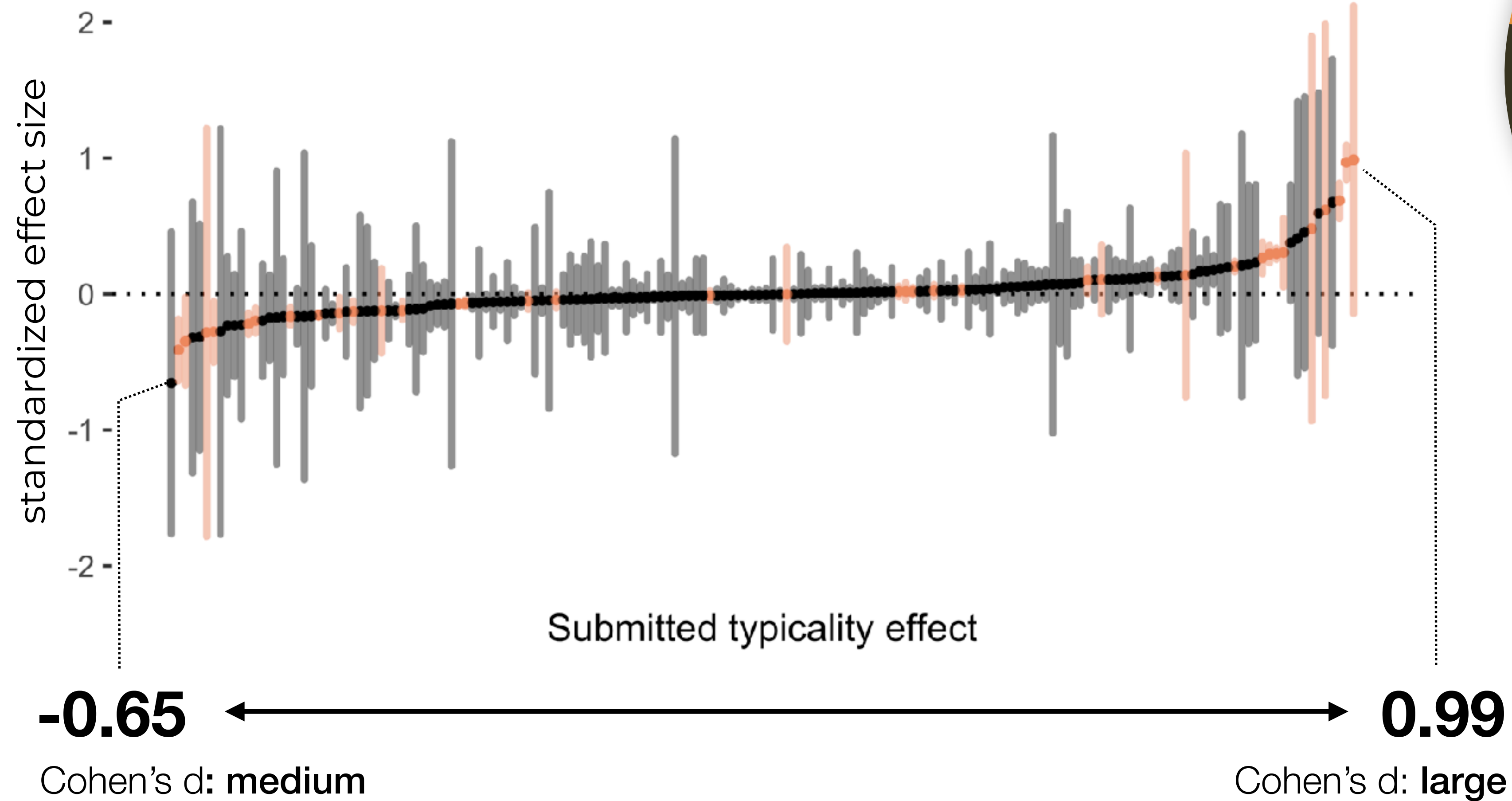


Do speakers acoustically modify utterances to signal atypical word combinations?

3 RESEARCH TEAMS, 3 DIFFERENT ANSWERS







Silberzahn et al. (2018)

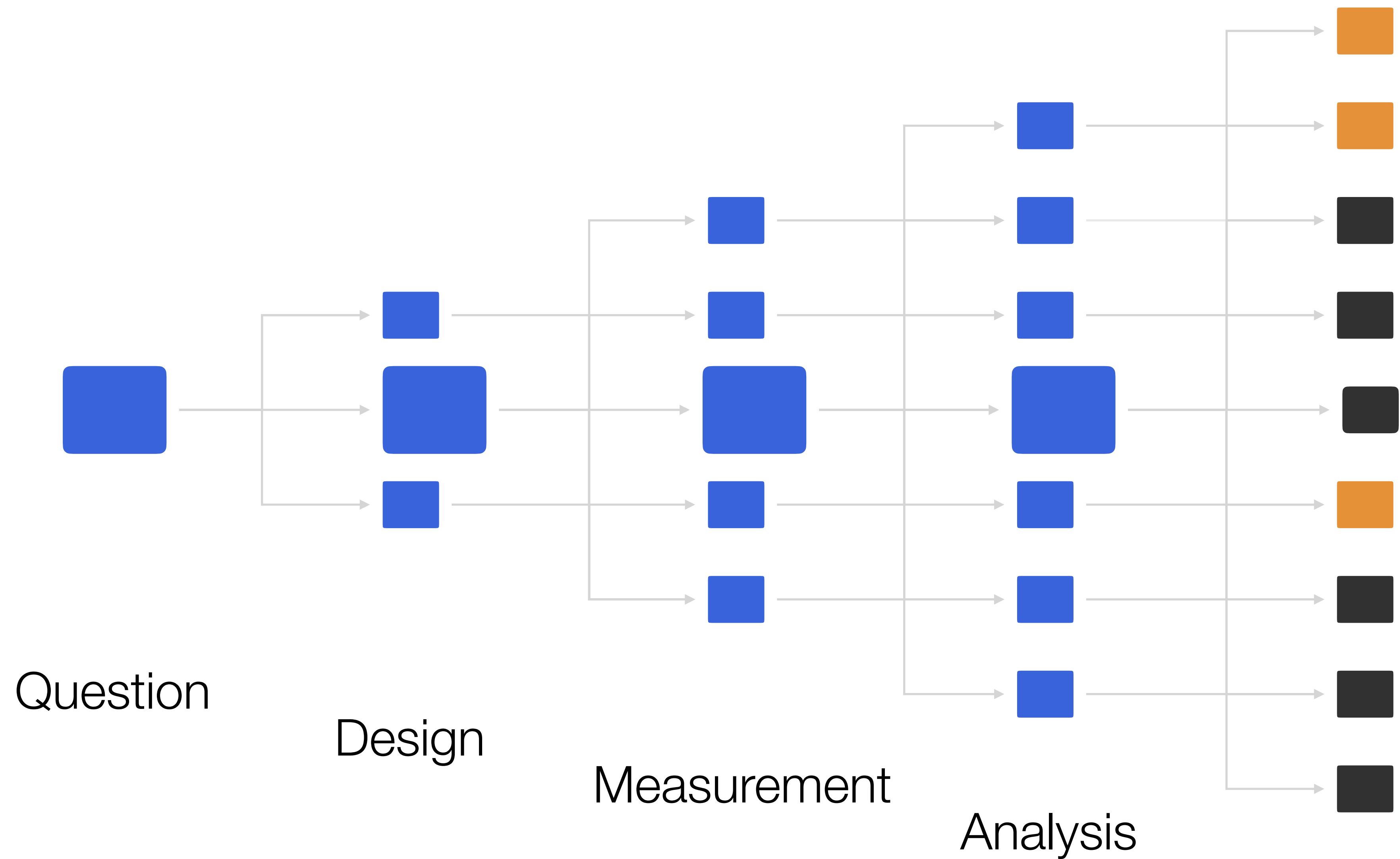
29 analyst teams

21 unique covariate combinations

69% found an effect

see also
Dutilh et al. (2019)
Starns et al. (2019)
Bastiaansen et al. (2020)
Botvinik-Nezer et al. (2020)
Parker et al. (2020)





THE PROBLEM: THEORETICAL VAGUENESS

e.g. Scheel (2021), Feest (2019), Meehl (1990)

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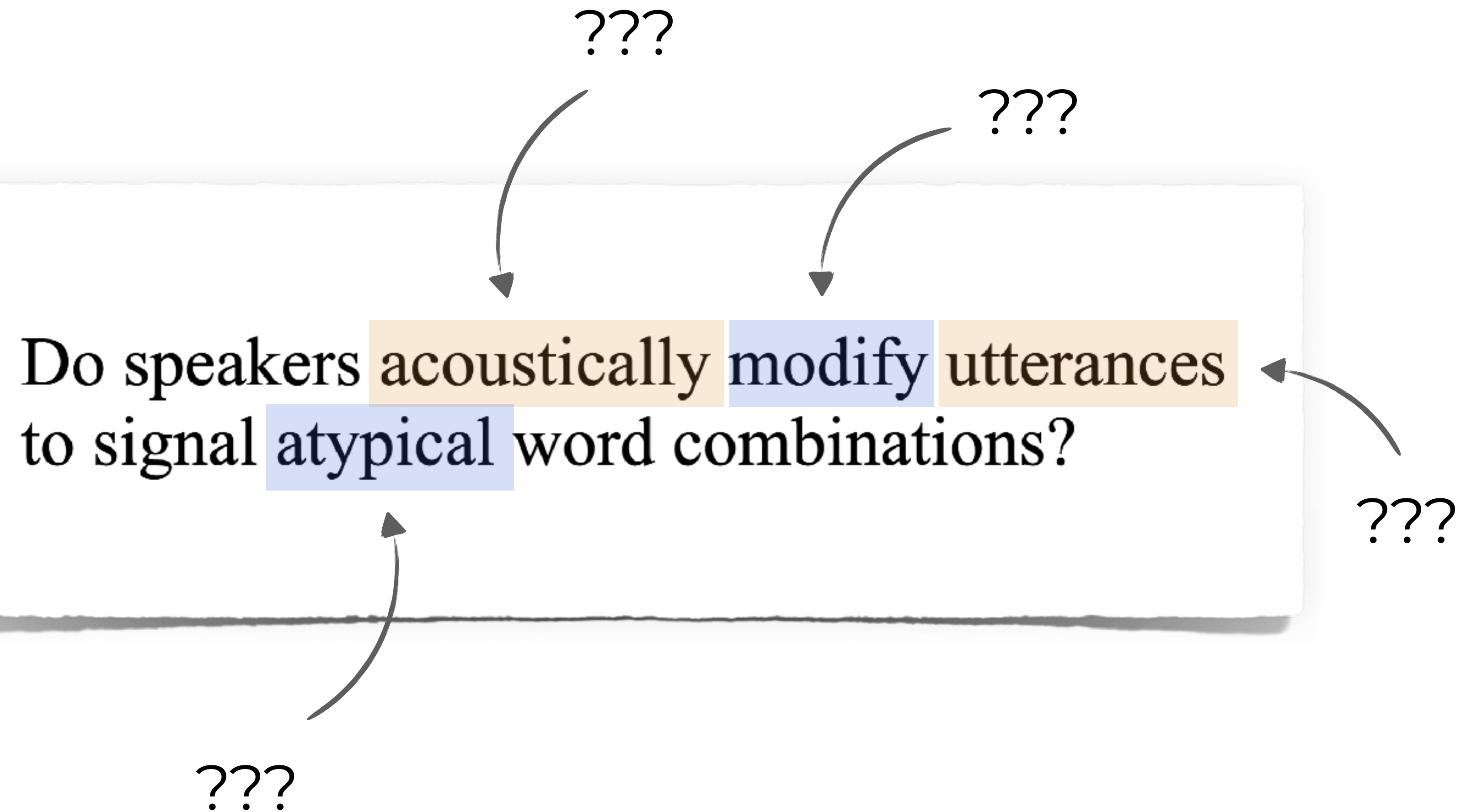
???

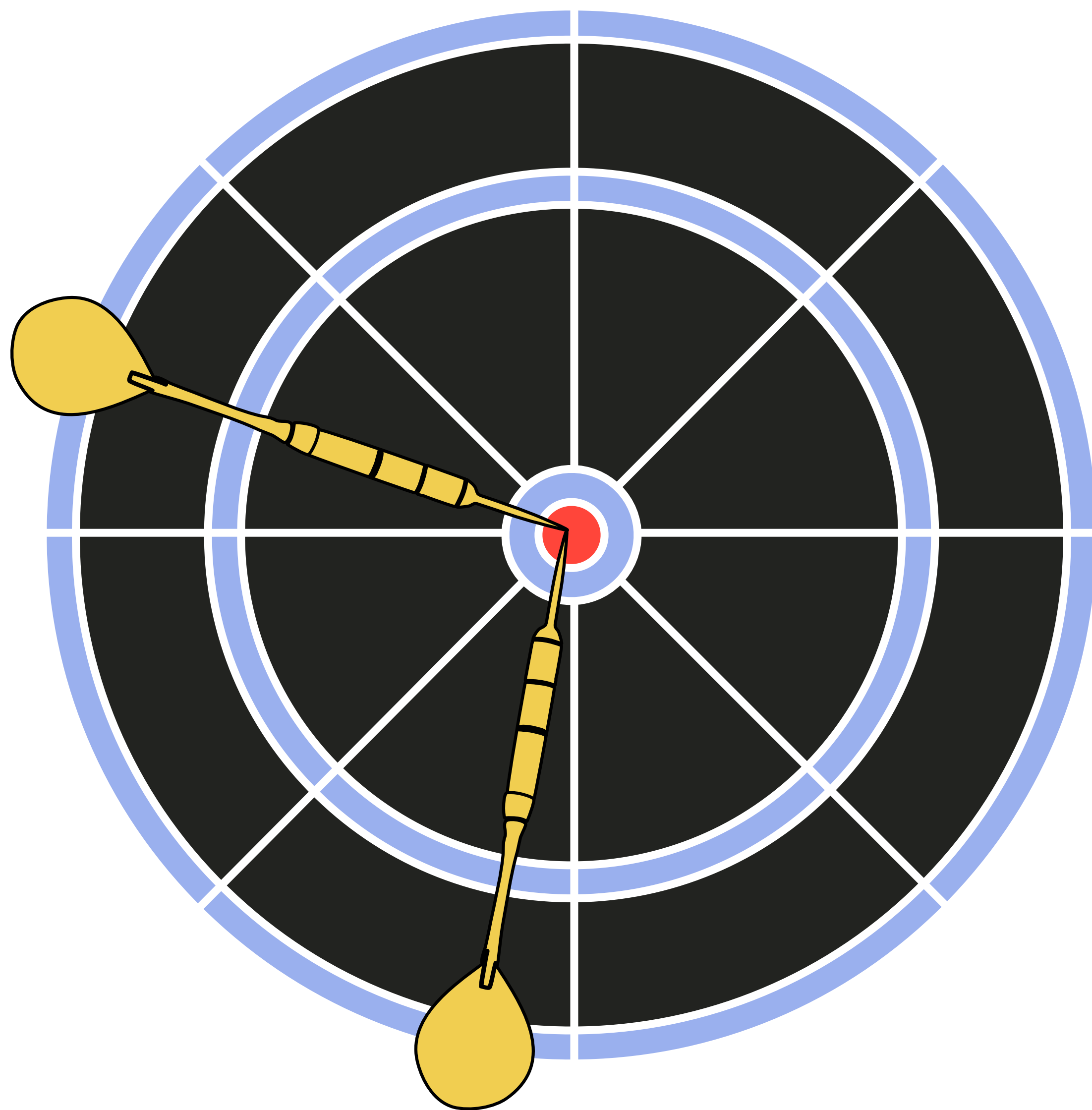


Do speakers **acoustically** modify utterances to signal atypical word combinations?

THE PROBLEM: THEORETICAL VAGUENESS

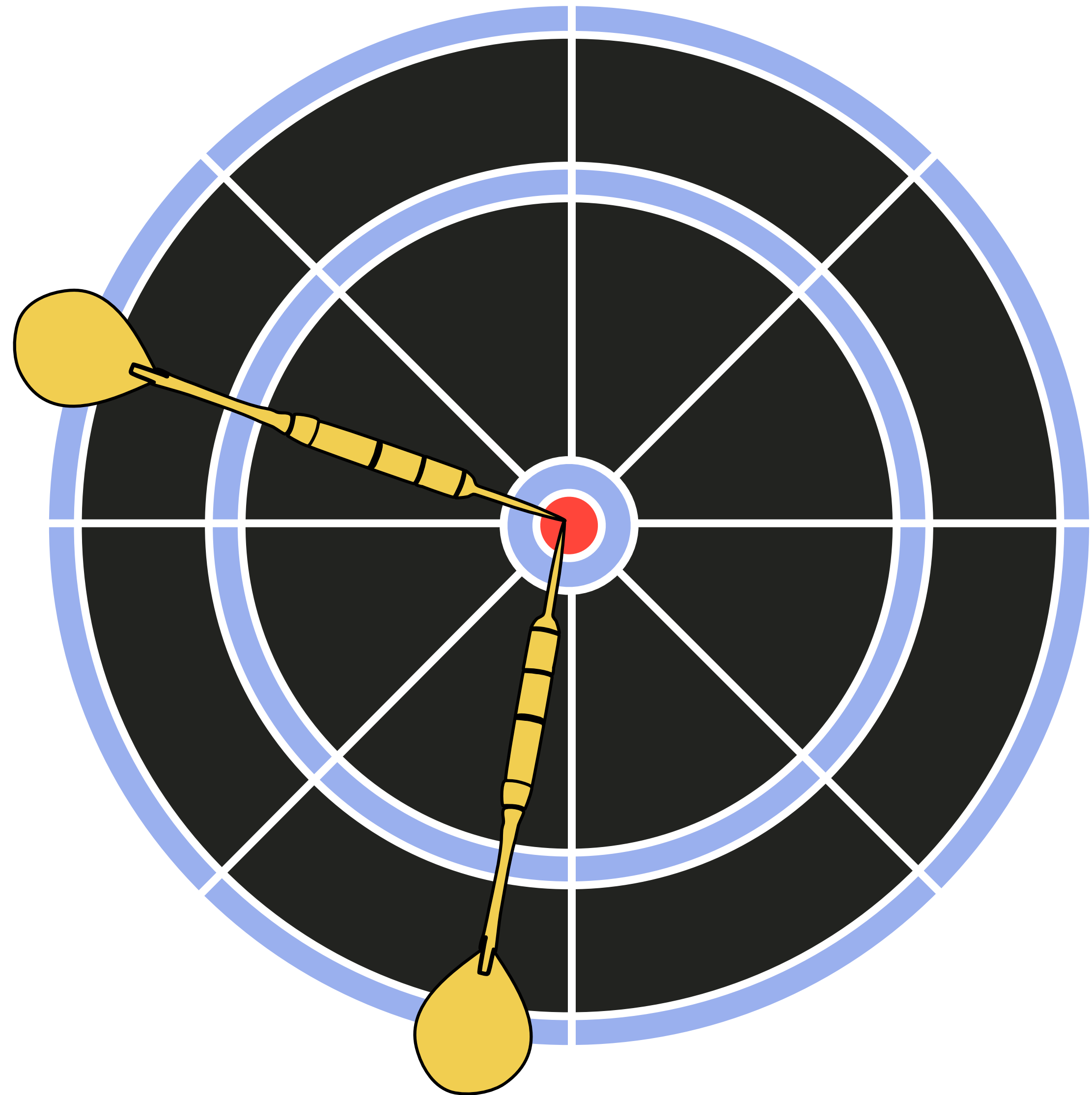
e.g. Scheel (2021), Feest (2019), Meehl (1990)





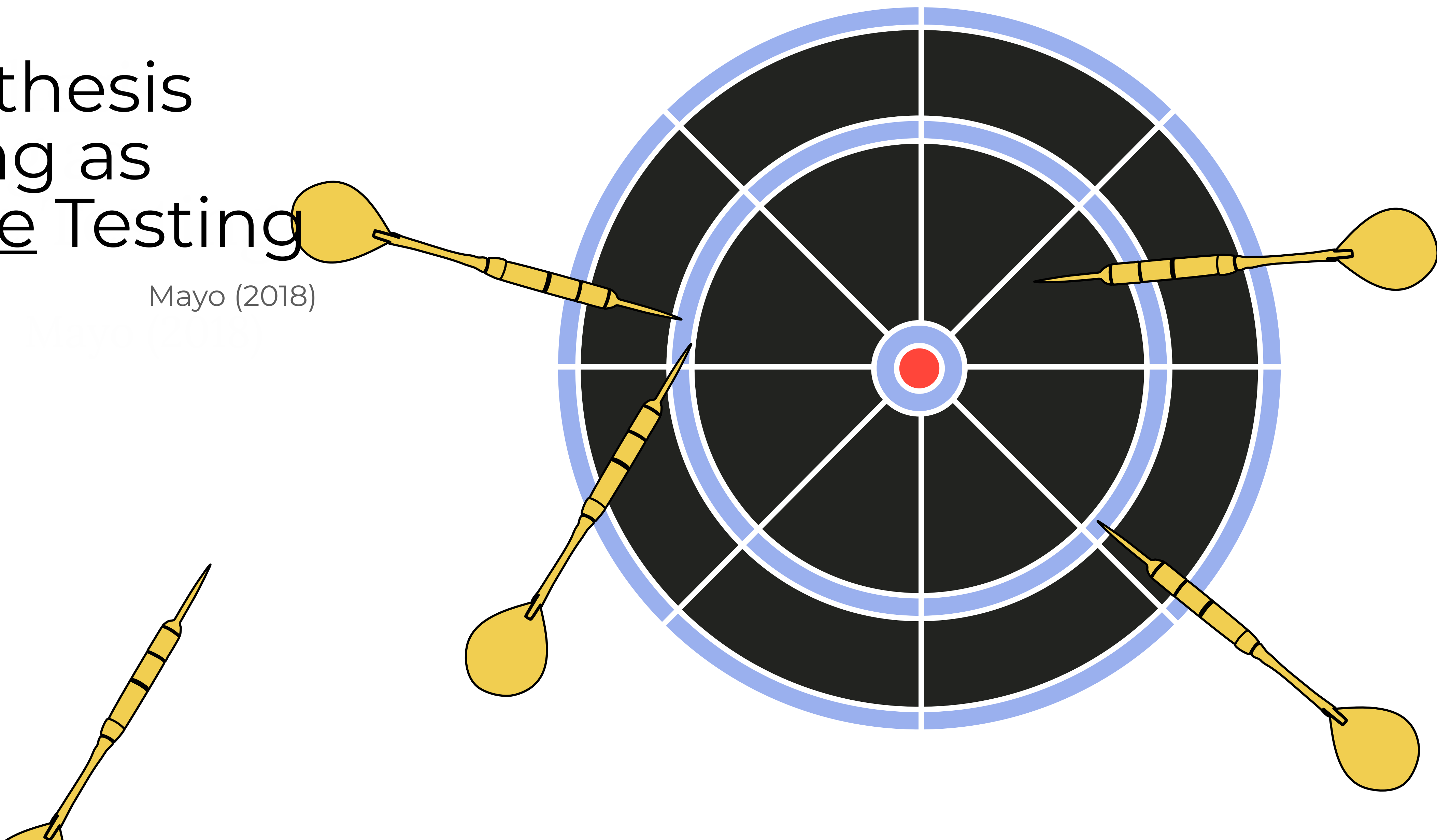
Hypothesis Testing as Severe Testing

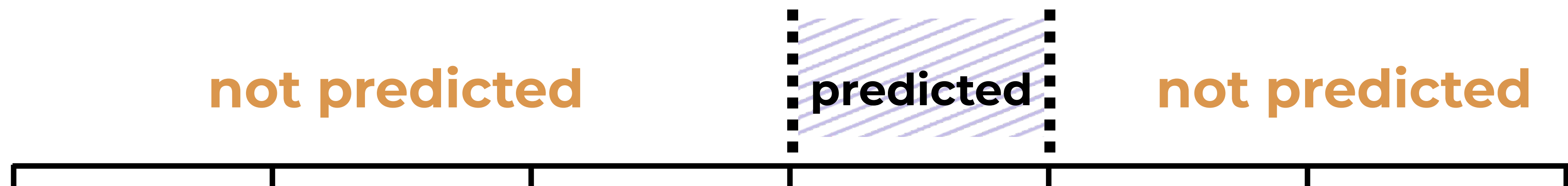
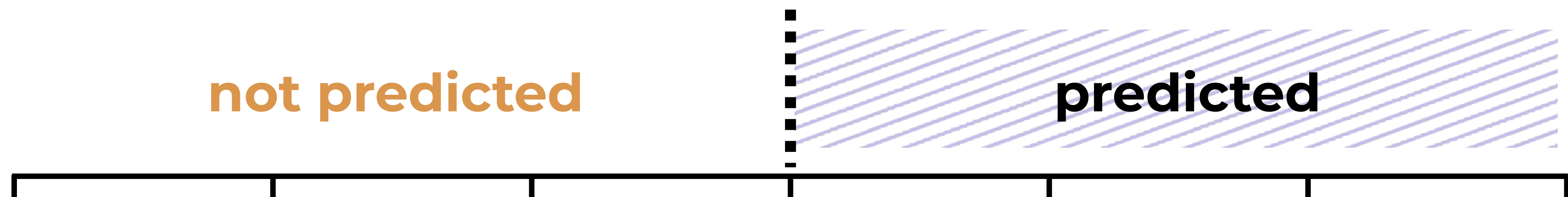
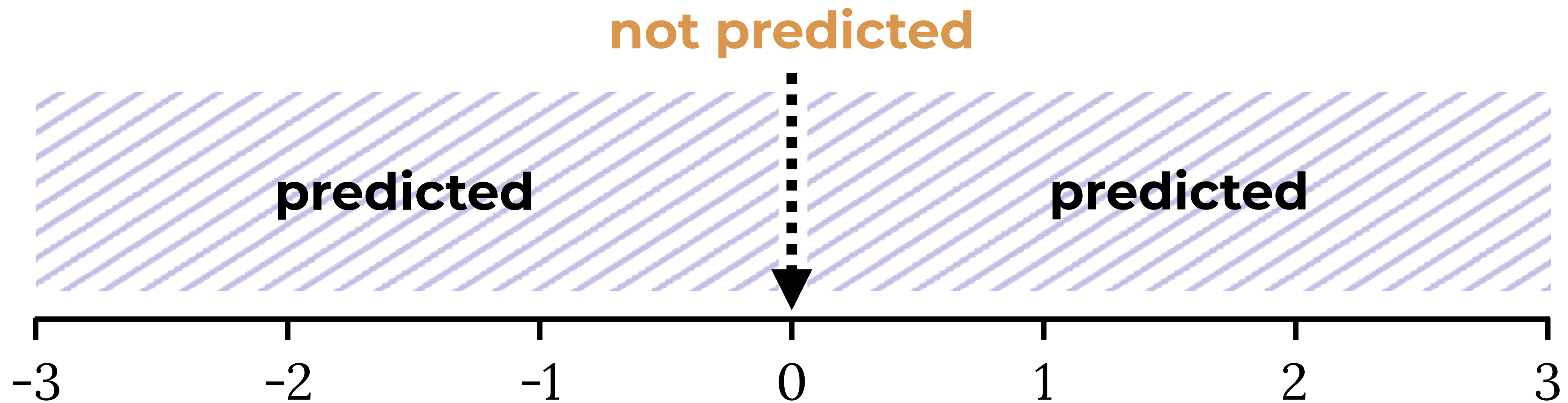
Mayo (2018)

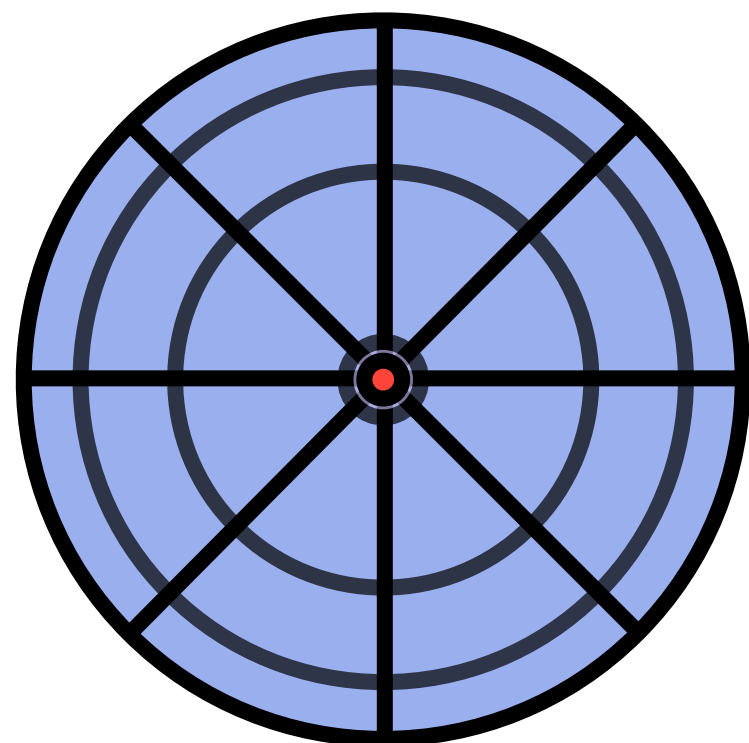


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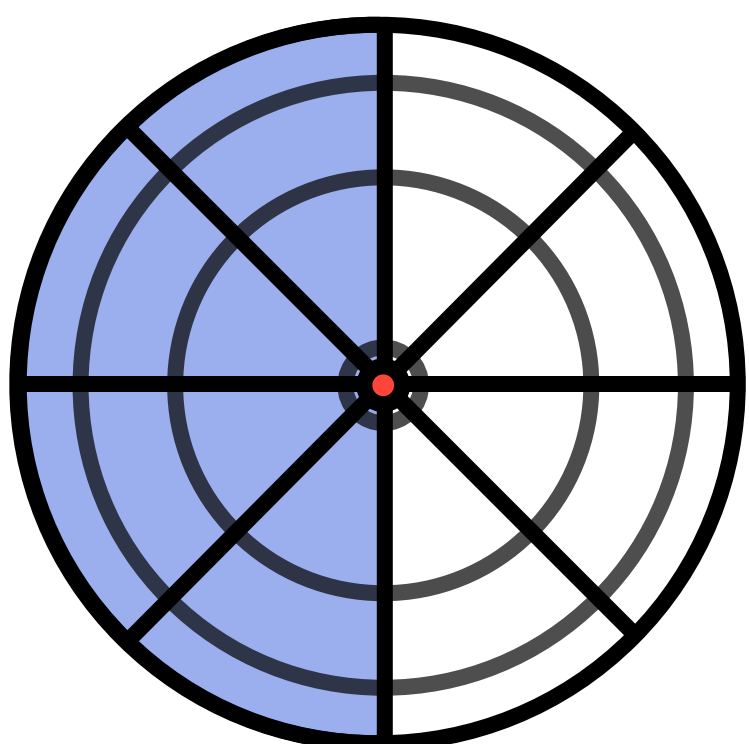
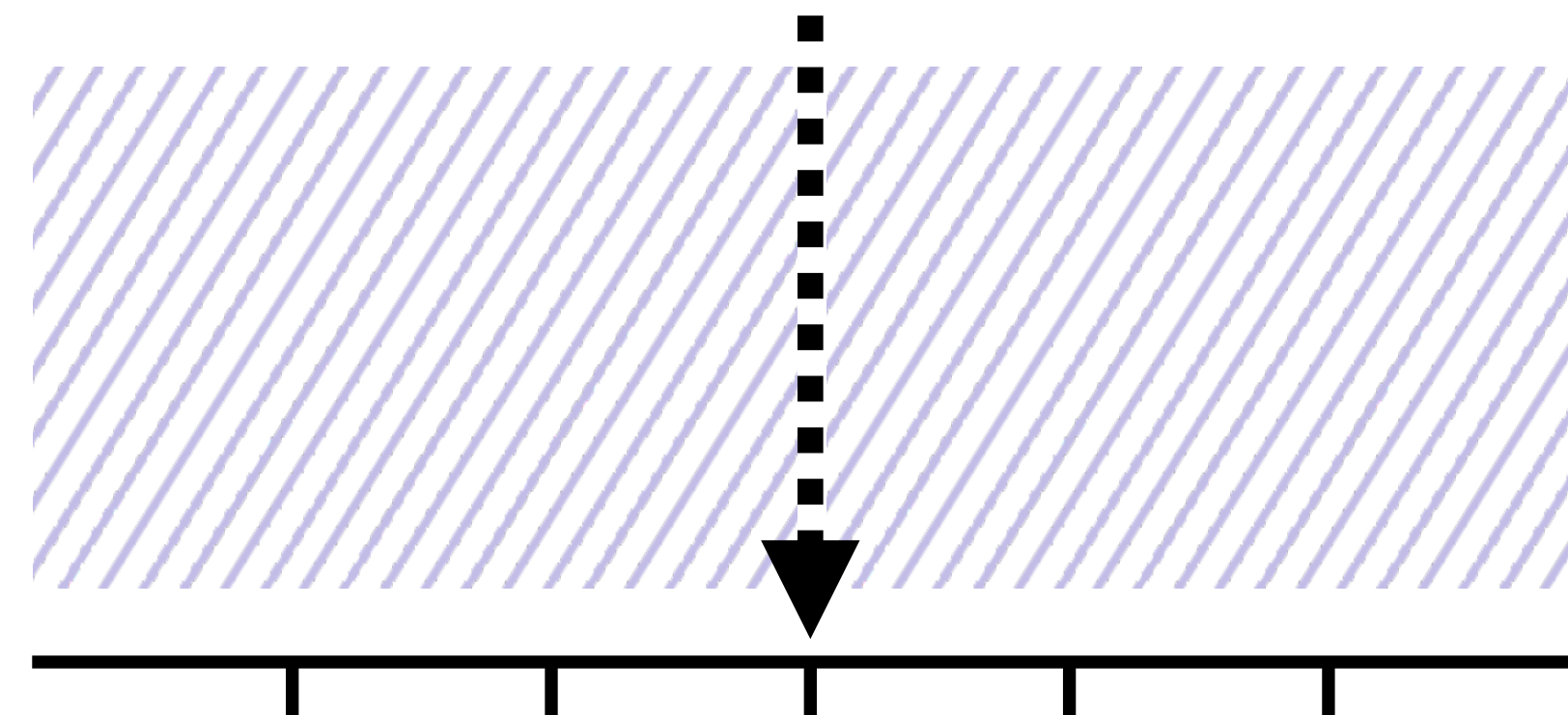
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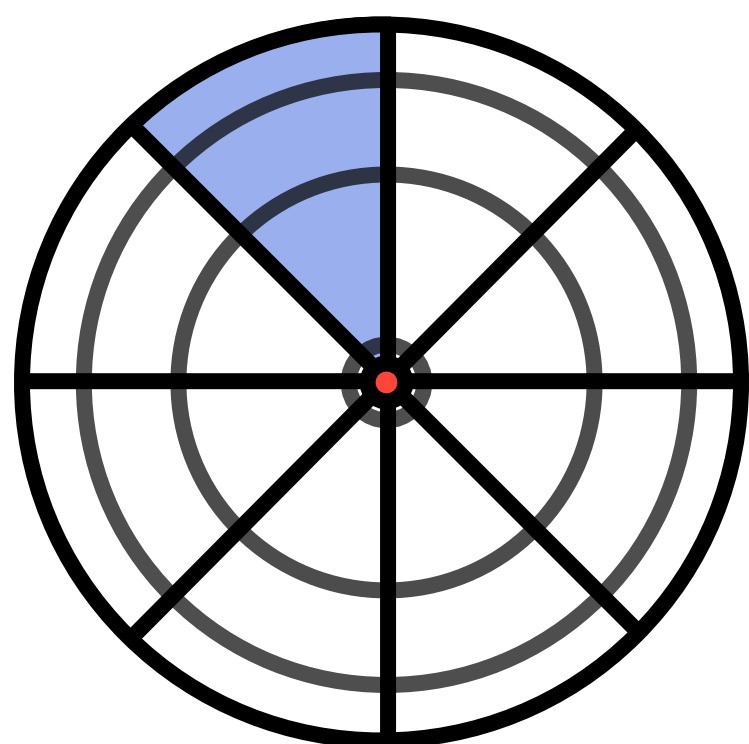
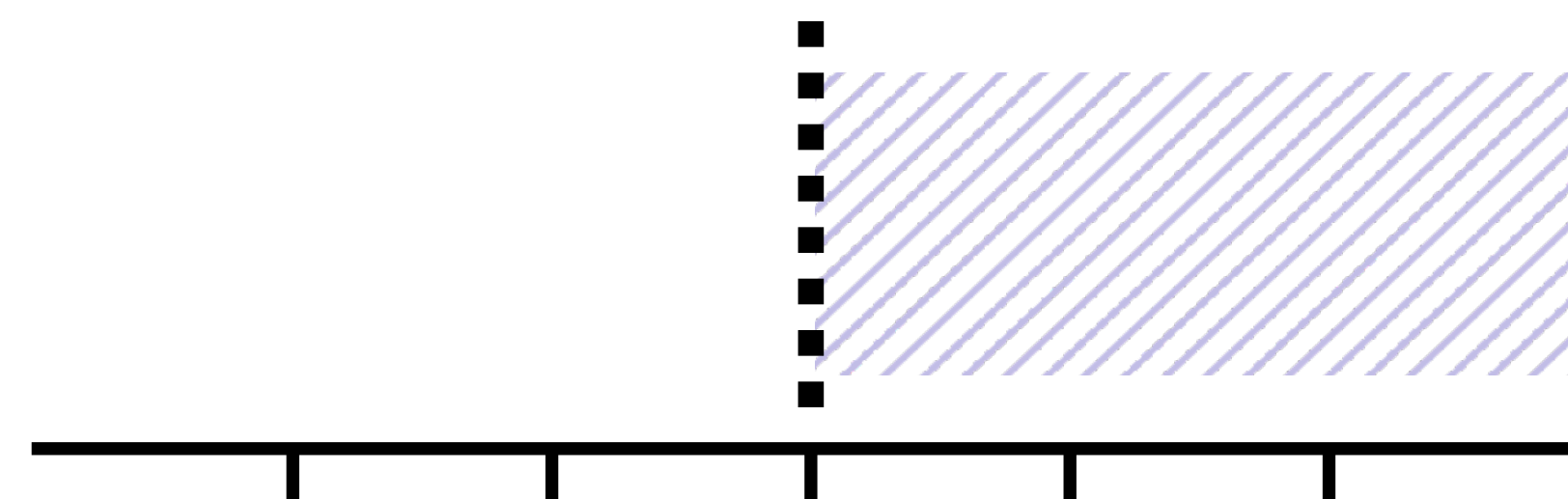




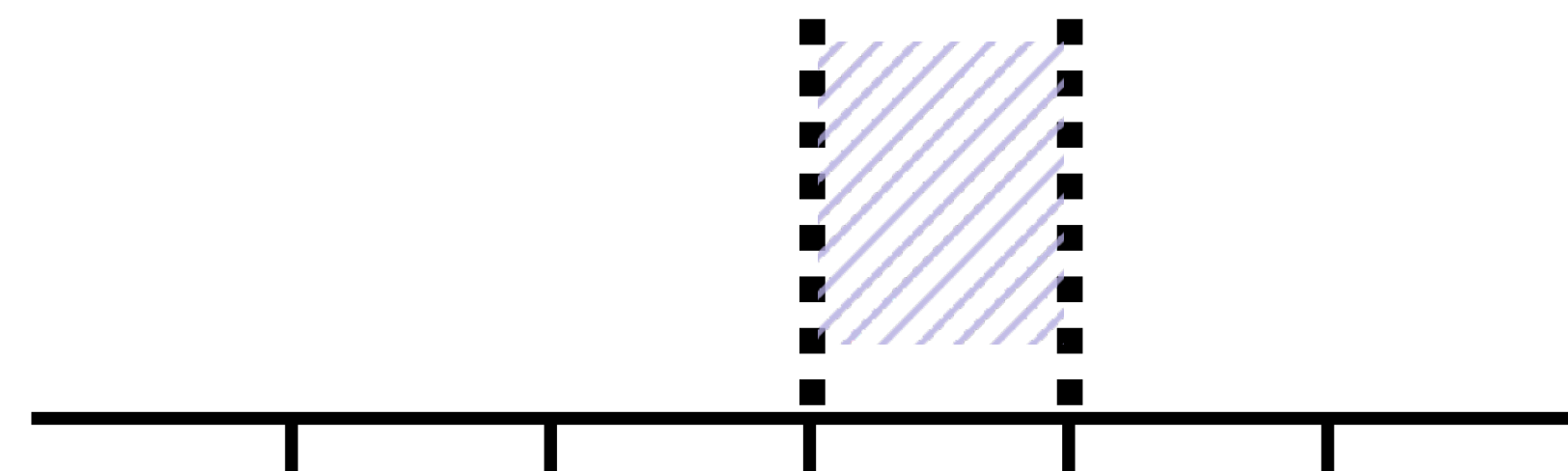
null-hypothesis

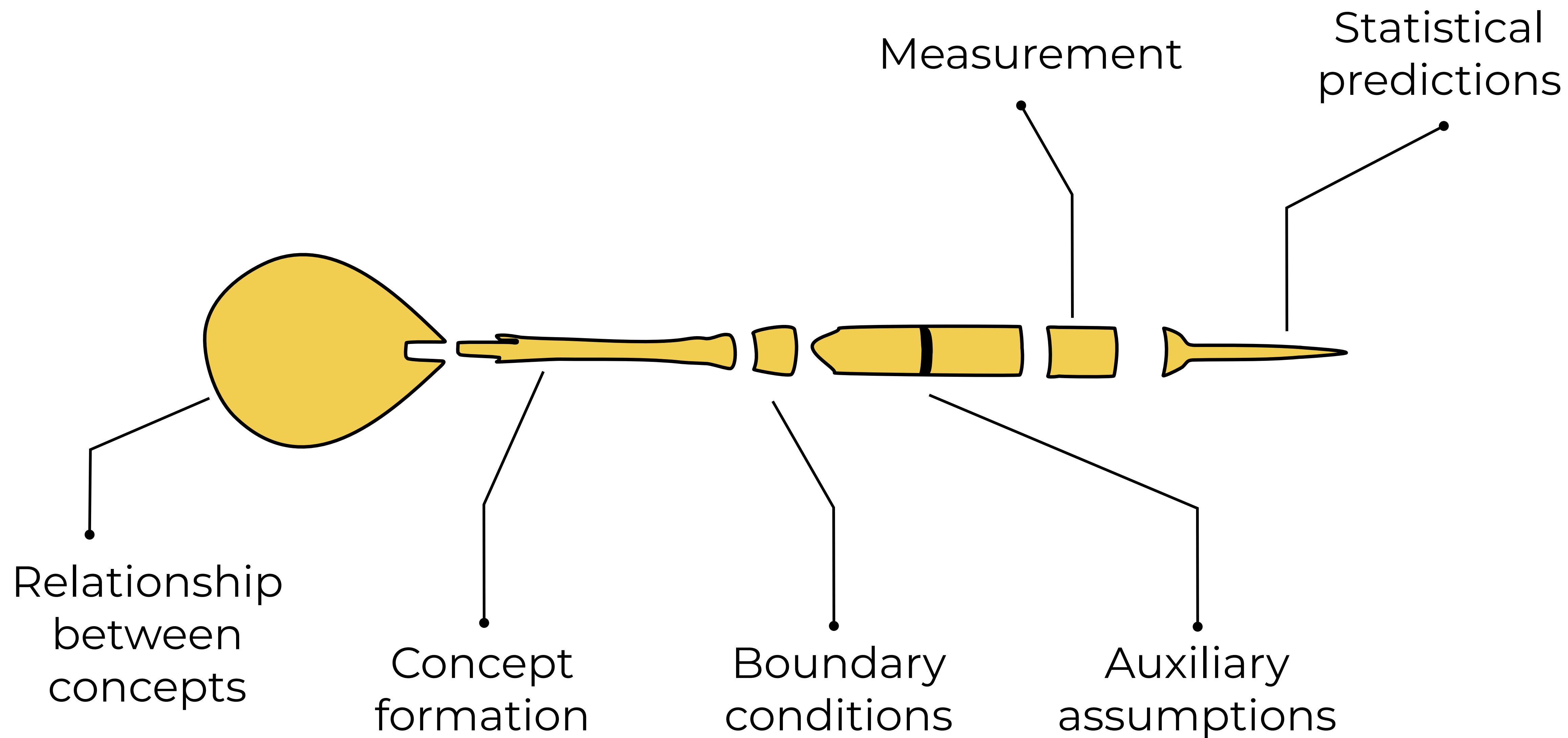


directional hypothesis



range hypothesis





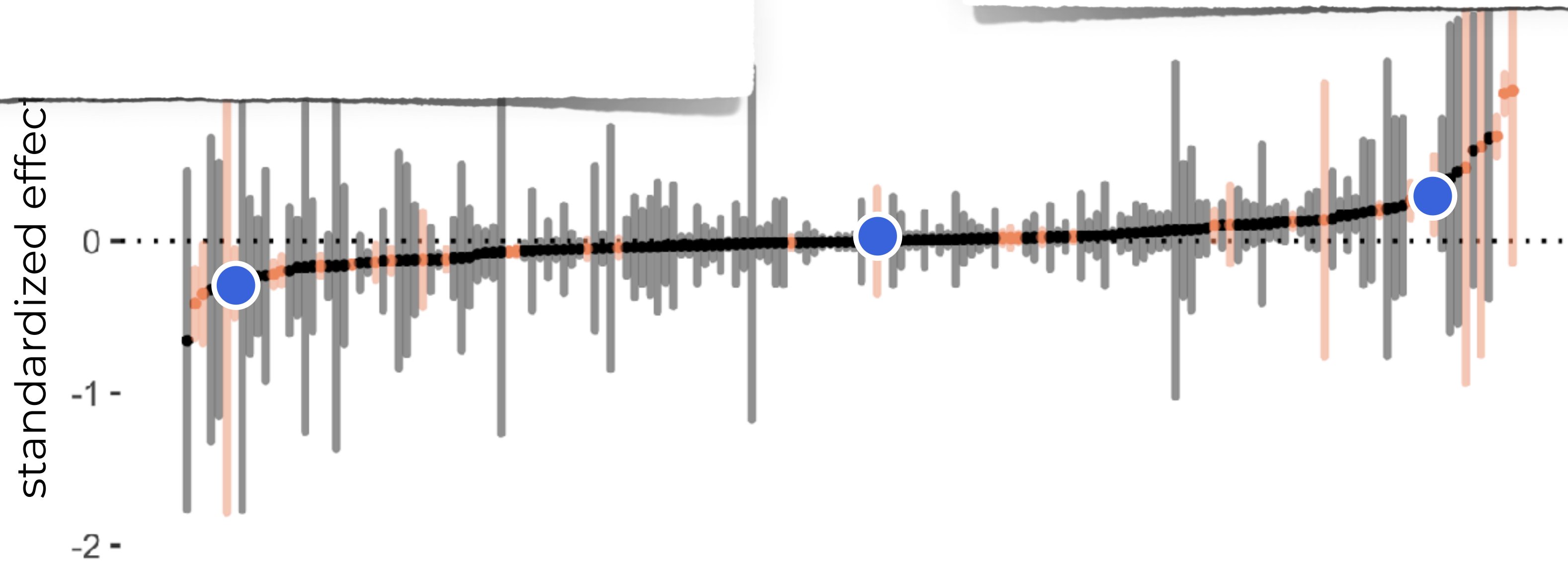
WE ARE NOT EVEN WRONG

e.g. Scheel (2021), Feest (2019), Meehl (1990)

Overweight is more prevalent in
children with immigrant background.

Increased risk of stillbirth after
flu infection.

Voice pitch is an acoustic
marker for schizophrenia.



POSSIBLE WAYS FORWARD

Better theory - Develop a better understanding how our theoretical (e.g. cognitive, social, structural) constructs are grounded in our measurement systems.

↳ One promising tool: **computational models**

van Rooij & Blokpoel (2020), Guest & Martin (2021), Scheel et al. (2021), Devezer et al. (2021)

More transparency - Make all decisions and their reasoning detectable for other to evaluate (e.g. share scripts, code, raw data, protocols)

Munafò et al. (2017), Vazire (2017), Roettger (2019), Bochynska et al. (2023)

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Epistemological humility - Adjust our (un)certainty in claims in light of the researcher degrees of freedom

Breznau et al. (2022)

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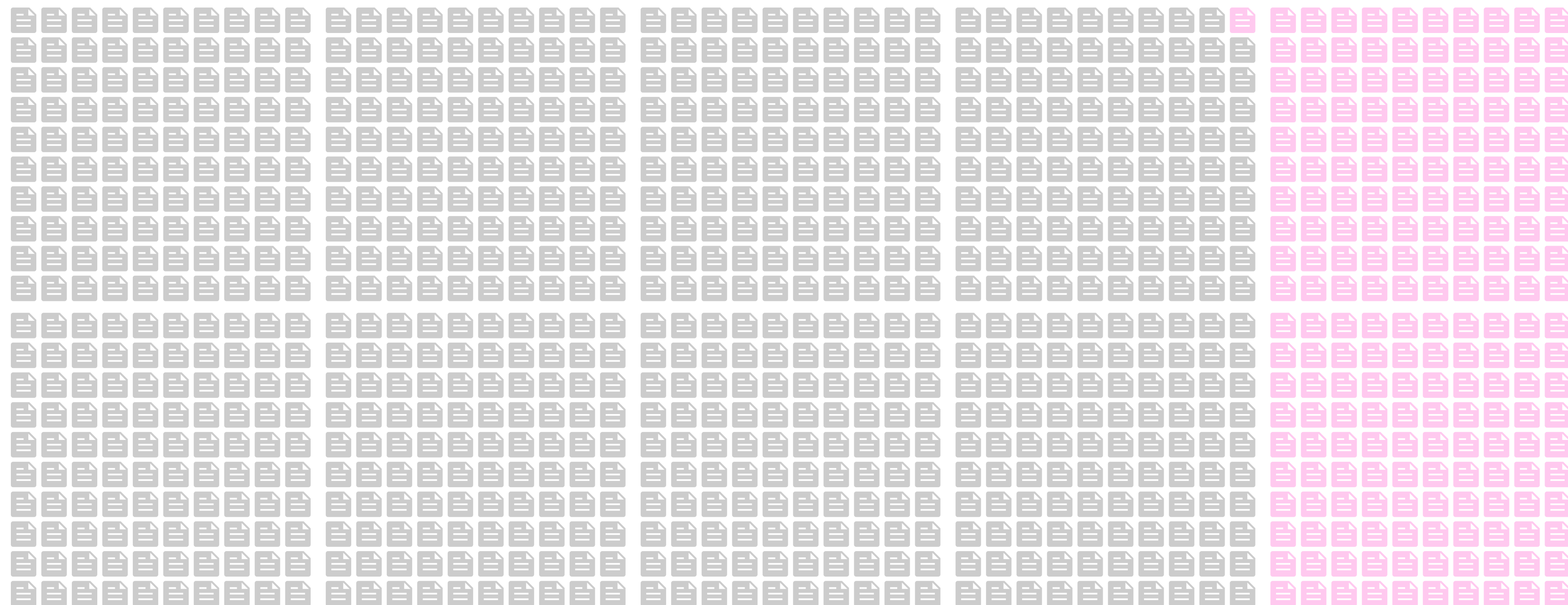
Epistemological humility - Adjust our (un)certainty in claims in light of the researcher degrees of freedom

THANK YOU

Get in touch
timo.b.roettger@gmail.com

HOW TRANSPARENT ARE WE?

Bochynska et al. (2023)

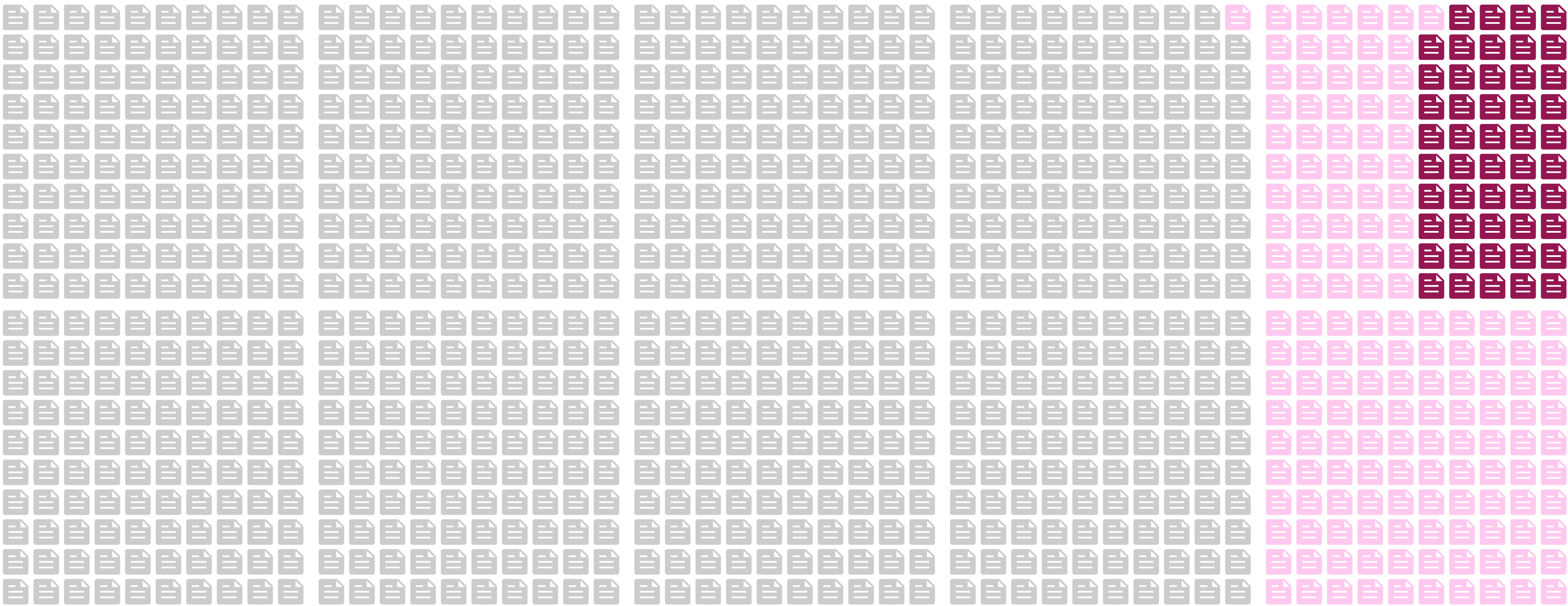


available

-  materials
-  raw data
-  processed
-  scripts

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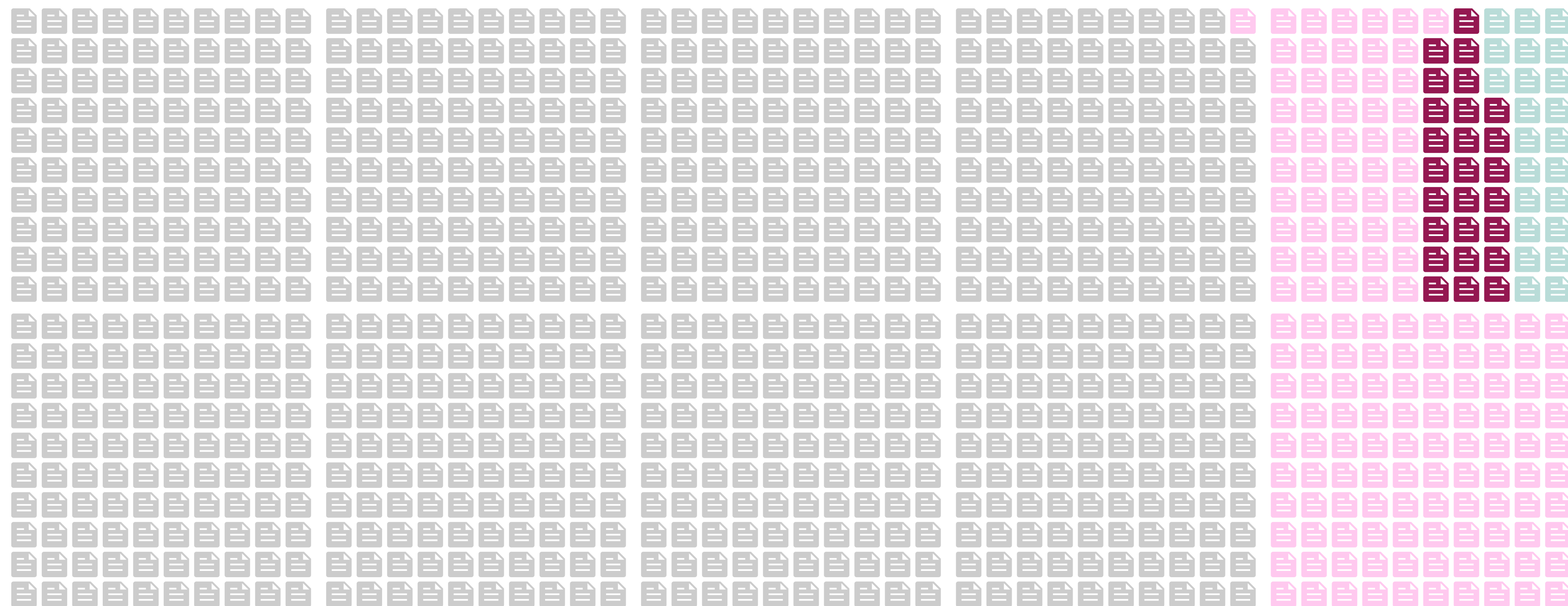


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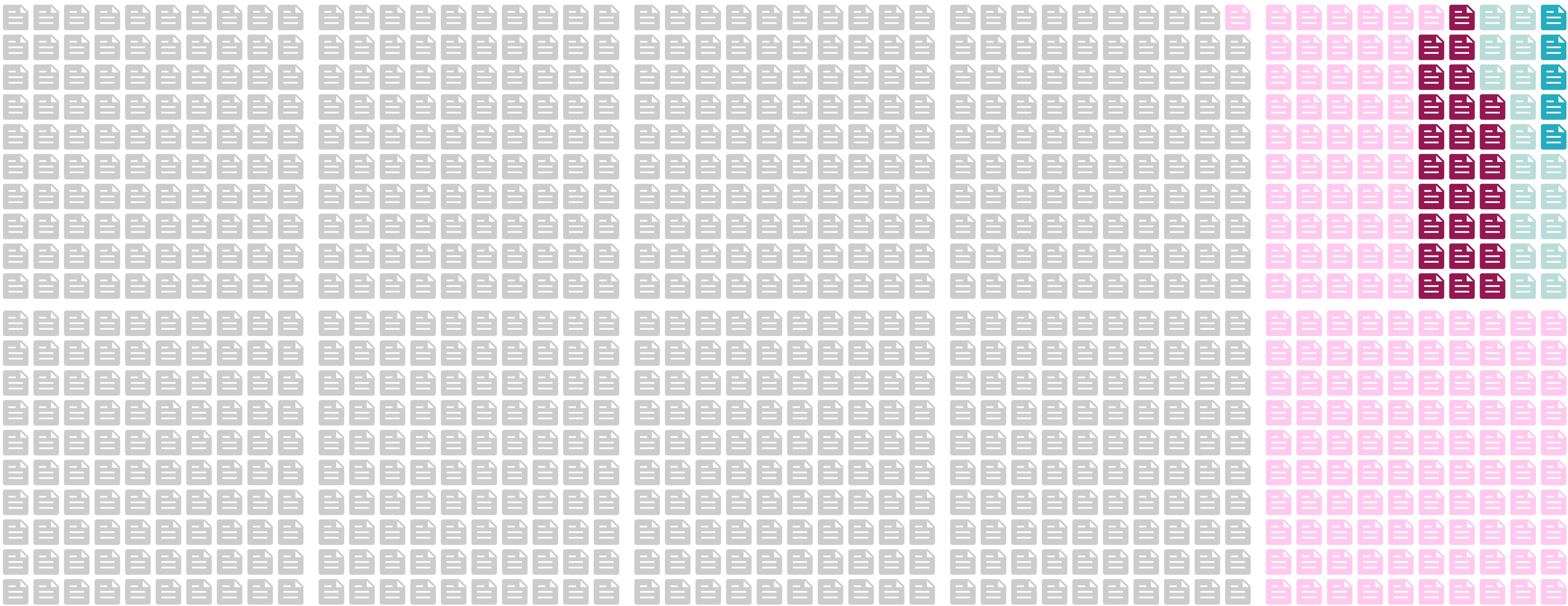


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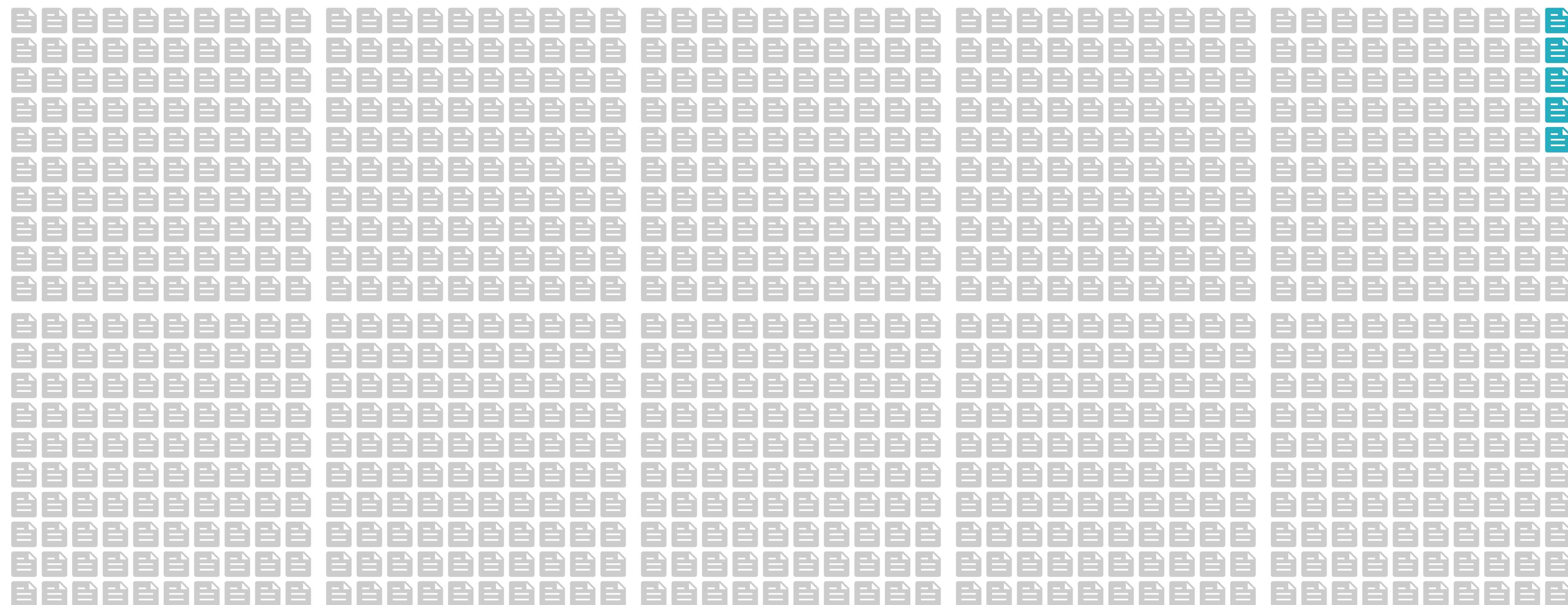


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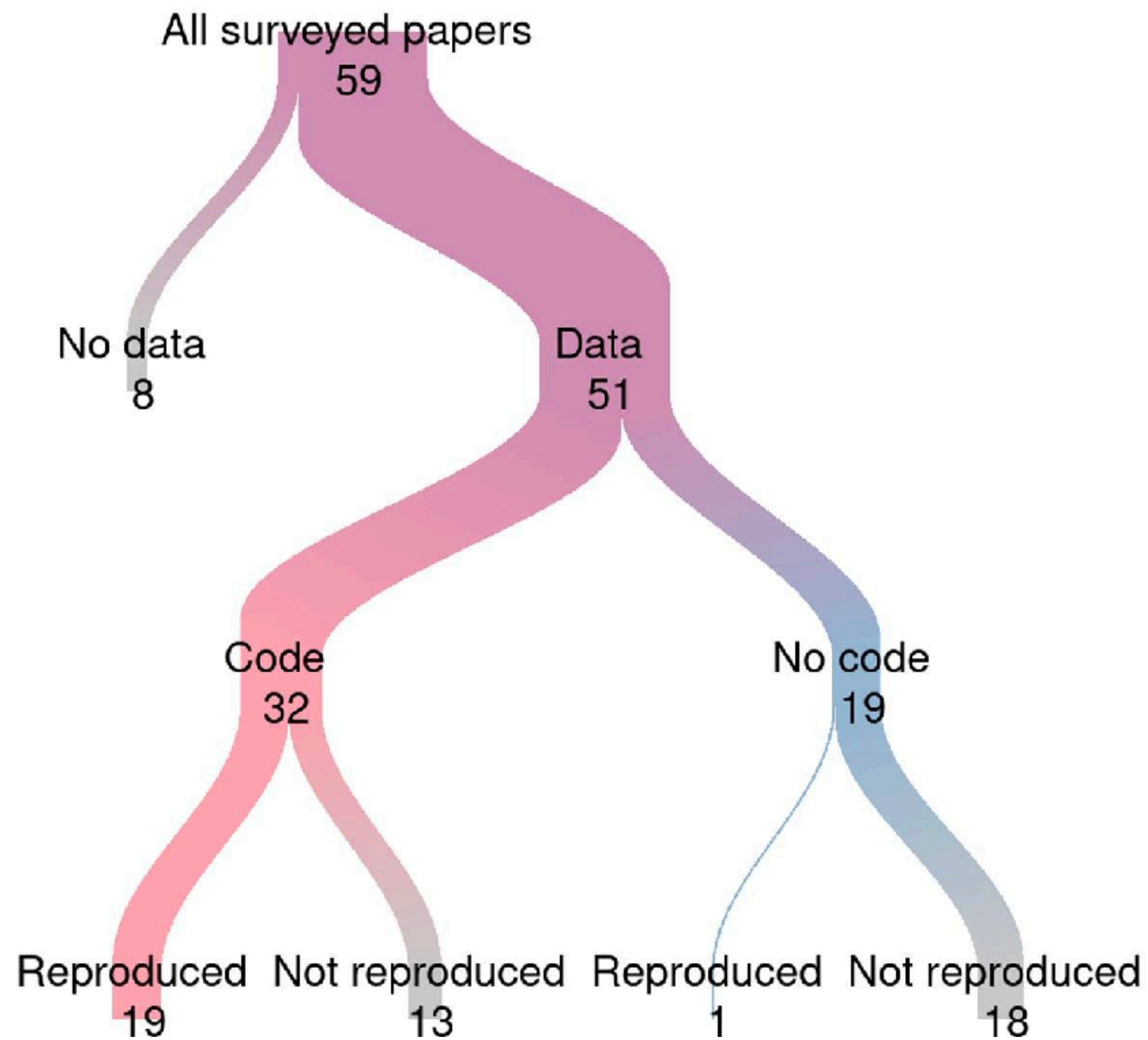
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HOW TRANSPARENT ARE WE?

Bochynska et al. (2023)



in principle
reproducible



**BUT
MANY ANALYSES
AREN'T
REPRODUCIBLE**

Laurinavichyute et al. (2022)

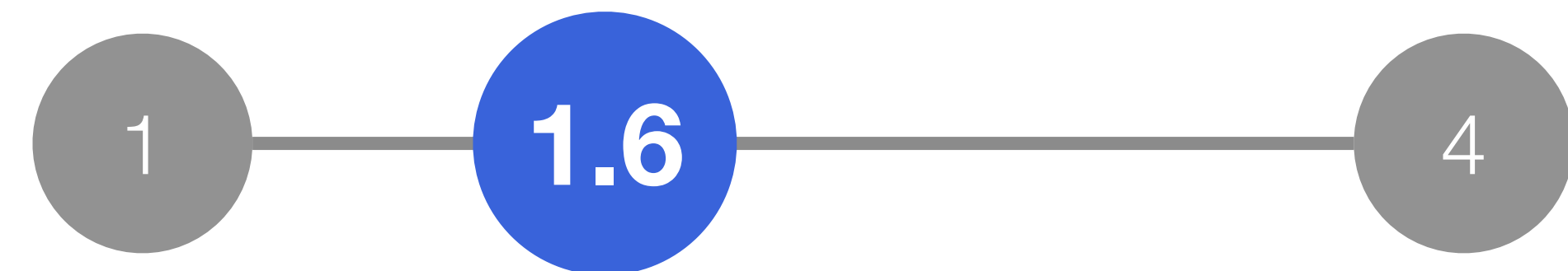
see also Hardwicke et al. (2018)

DIFFERENCE BETWEEN STATISTICAL ANALYSES

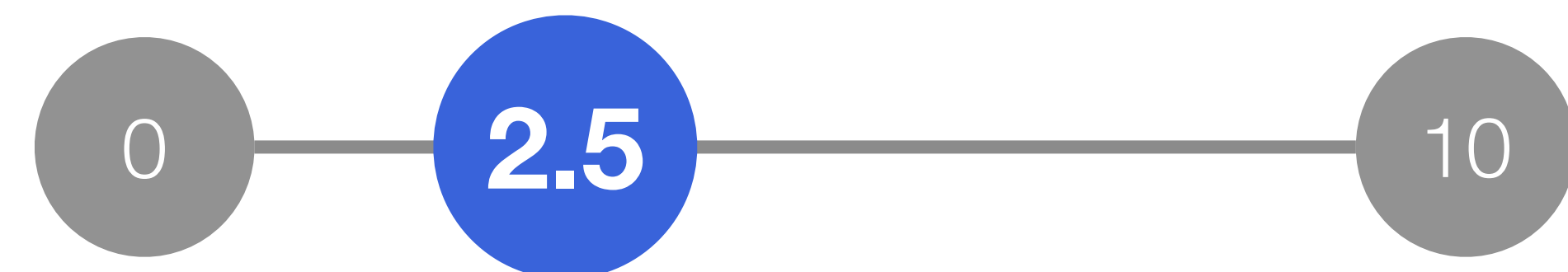
submitted models



predictors



random effects



minimum

mean

maximum

DIFFERENCE BETWEEN STATISTICAL ANALYSES

submitted models



predictors



information str.

trial #

semantics

FOS

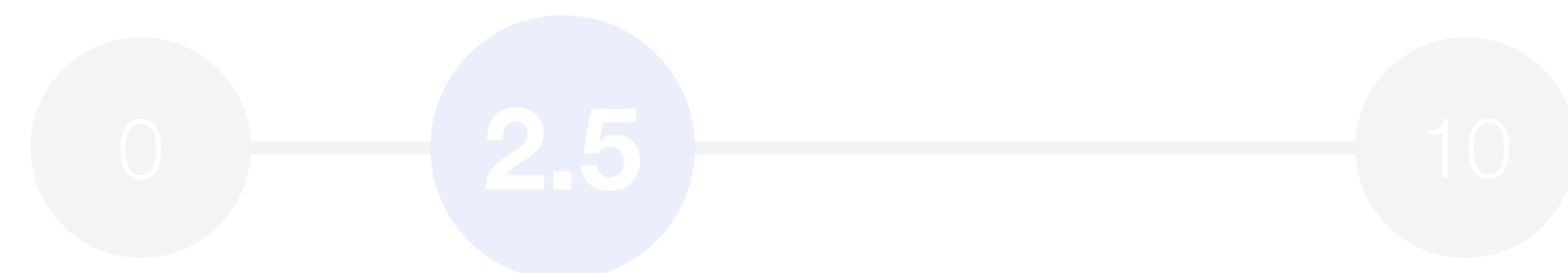
vowel quality

syllable str.

speaker gender

speech rate

random effects



minimum

mean

maximum